

Multi-Horizon Failure-Risk Models for Battery Failure Prediction: How Much Cross-Chemistry Transfer Is a State-of-Health Shortcut?

Anonymous

Affiliation 1, City, Country (e-mail: author1@example.com)

Abstract

Predicting whether a lithium-ion cell will fail inside a short operating window is a different question from estimating its remaining useful life, and it is the one that matters most for real-time dispatch in electric vehicles and grid storage. This study widens a recently proposed multi-horizon failure-risk classification framework, first shown on one model and one dataset, into a validity-focused evaluation covering four model families, a true discrete-time hazard model, trivial health-state baselines, and six transfer targets spanning four chemistries — LCO, LFP, nickel-rich NCA and NMC — drawn from six laboratories, all under cell-disjoint splits and a leakage-clean, cross-fitted calibration protocol. Within-dataset discrimination is reliable, with fold-mean area under the curve (AUC) of 0.90–0.96 across the tree ensembles under Platt calibration.

The central result is a controlled State-of-Health (SOH) ablation: apparent cross-dataset transfer with SOH included (pooled AUC up to 1.00) collapses without SOH (at best 0.58 at $H=20$), and much of the with-SOH signal is reproducible by a distance-to-threshold rule on SOH alone (0.912 AUC on the 141-cell MIT–Stanford Severson set). Because SOH also enters the failure definition and the label window includes the current cycle, all with-SOH numbers are diagnostic-proximity upper bounds rather than pure prognosis: relabeling failures by voltage sag alone, a criterion SOH does not define, cuts with-SOH transfer by 0.17 (0.90 to 0.72 pooled on Severson at $H=20$), and restricting evaluation to rows scored while SOH is still above 0.80 removes the concurrent-detection component everywhere. Same-chemistry cross-laboratory controls confirm that the collapse is driven by dataset shift at least as much as by chemistry, so SOH’s dominance is

a dataset- and chemistry-dependent health-state shortcut. The evaluation is replicated on 60 new cells of NMC, NCA and LFP chemistry from two further laboratories (the public Battery Archive): with-SOH transfer spans 0.77–1.00 pooled AUC and ranges down to 0.36–0.91 without SOH outside the single-condition HNEI group (paired deltas 0.06–0.64), and no competing shortcut hides in temperature or C-rate. Isotonic calibration largely fails to transfer, and the three tree ensembles deploy on a \$12 ESP32-S3 in under a millisecond per inference for a single SRAM-resident model (0.76–2.36 ms from flash for the full binary).

Keywords: Battery failure prediction, multi-horizon failure-risk classification, state-of-health shortcut, cross-chemistry transfer, dataset shift, probability calibration, discrete-time hazard model, SHAP, embedded machine learning, ESP32-S3, battery archive.

I. Introduction

Lithium-ion batteries sit underneath the electrification of transport and the decarbonization of grids, yet their long-term reliability remains hard to certify in the field. Conventional prognostics casts the problem as remaining-useful-life (RUL) regression, in which a model predicts a continuous cycle count to end of life [1, 2]. RUL regression is appropriate for capacity-fade studies, but it answers a question operators rarely ask in real time: a battery management system needs to know whether a specific cell will survive the next mission, not how many cycles it has left in aggregate. Predicting lifetime from field data rather than laboratory cycling remains an open challenge [3]. Review studies group prior methods into physics-based, purely data-driven, and hybrid families [1]. Tree ensembles, such as Random Forest [4], XGBoost [5], and LightGBM [6], have been applied to capacity and RUL estimation, but their use for binary failure-risk classification at multiple horizons is recent [7].

Shikdar and Laaksonen [7] recast the problem as multi-horizon failure-risk classification. For a given cycle t and horizon H , the model predicts whether failure, defined as SOH below 0.80 or average voltage sag beyond 6%, occurs within the next H cycles. Their histogram-based gradient boosting model on the NASA 18650 set (37 LCO cells) reached AUC of 0.87–0.90 across $H \in \{10, 20, 30, 50\}$. That work left three gaps open. First, one model on one dataset says nothing about sensitivity to the learner, the hyperparameters,

or the calibration method. Second, no cross-dataset check existed, so nothing stopped a model trained on LCO cells from being trusted on LFP cells. Third, no path existed from a Python research prototype to the microcontroller where a battery management system actually runs.

This paper begins from those gaps but is organized around a sharper validity question that emerged while closing them: *when a failure-risk model appears to transfer between datasets and chemistries, is anything about degradation being transferred, or is the model merely reading a health-state variable that encodes proximity to the failure criterion itself?* The question matters because the composite failure label is defined from SOH, and SOH is also the strongest available feature. A model can therefore score well on transfer while learning “distance to the SOH = 0.80 threshold,” a quantity that says nothing about chemistry-specific degradation physics. A hostile reading of any cross-chemistry result that ignores this confound is, in our experience, correct more often than not.

Two adjacent literatures bear on these questions. Transfer learning for battery state estimation is a growing area: Sahoo et al. [8] fine-tuned an SOH estimator across NASA and CALCE cells, and Lu et al. [9] pretrained on large-scale degradation data before adapting to new cell types. Both target regression-based SOH estimation rather than classification-based failure risk, and the classification setting adds two complications: the label depends on the chosen horizon H , and the calibration of probability outputs, which decisions depend on, may degrade under distribution shift. On calibration, Platt scaling [10] fits a sigmoid through logistic regression on the model’s raw scores, while isotonic regression [11] fits a non-decreasing step function. Niculescu-Mizil and Caruana [12] showed that Platt is safer on small datasets and isotonic can overfit, and Huang et al. [13] extended this to imbalanced classification; no prior study has compared the two calibrators under cross-chemistry distribution shift in battery prognostics. A companion robustness study further found that under realistic operating perturbation the calibration layer is the primary point of failure, and that recalibration on a small sample of the operational data largely restores calibration [14]; whether the same repair survives a cross-chemistry shift is examined in this work. On explainability and deployment, TreeSHAP [15] provides consistent feature attributions and has become the standard tool for diagnosing ensemble inputs, while edge inference removes latency, connectivity requirements, and privacy concerns [16]. Online recalibration methods for shifting score distributions are also being studied [17]. Table 1 situates this work against these

Table 1: Comparison of Representative Battery Studies With This Work

Study	Family	Data	Position relative to this work
Meng and Li [1]	Prognostics review	Various	Groups methods into physics-based, data-driven, and hybrid families
Breiman [4], Chen and Guestrin [5], Ke et al. [6]	Tree ensembles	Capacity, RUL	Established ensembles for RUL regression, not failure-risk classification
Sahoo et al. [8]	Transfer learning	NASA, CALCE	Transferred SOH regression across chemistries; regression only
Lu et al. [9]	Deep learning	Large SOH corpora	Pretraining for SOH estimation; regression only
Platt [10], Zadrozny and Elnan [11], Nielsen-Mizil and Caruana [12], Huang et al. [13]	Probability calibration	Various	Calibrators compared in-domain; no cross-dataset setting
Lundberg et al. [15]	Explainability	Various	TreeSHAP attributions for tree ensembles
Gresik and Mrozek [16]	Edge computing	Various	Edge inference review without battery failure use cases
Gupta and Ramdas [17]	Calibration	Various	Online Platt scaling under distribution shift
Shikdar and Laaksonen [7]	Failure-risk classification	NASA	Single model, single dataset; baseline of this paper
This work	Failure-risk classification + validity analysis	NASA, CALCE, Oxford, Severson, Battery Archive NMC/NCA/LFP	Multi-model benchmark, SOH-shortcut diagnosis with label-, feature-, and dataset-level controls, discrete-time hazard model, calibration under shift, embedded deployment

lines of research.

The objective of this study is to close the three gaps identified above and, in doing so, to answer the validity question rigorously. The contributions are:

- a four-model benchmark, three tree ensembles (XGBoost, LightGBM, and Random Forest) plus a GRU sequence classifier, run on two LCO datasets (NASA 18650 and CALCE LCO/CX2) under cell-disjoint Group-KFold and leave-one-cell-out protocols with cross-fitted, leakage-free calibration;
- a true discrete-time hazard formulation, in which per-offset hazards are pooled into a survival product that is monotone in the horizon by construction, evaluated against the four independent fixed-horizon classifiers, whose predicted risks violate the corresponding monotonicity requirement on 32–45% of within-dataset rows (Table 13);
- a shortcut diagnosis of cross-dataset, cross-chemistry transfer: minimal baselines (a distance-to-threshold rule, SOH-only, cycle-only, and sensor-only logistic regressions), a factorial feature ablation over {SOH, cycle index, sensors}, a failure-definition ablation that relabels failures by SOH or by voltage sag alone, and same-chemistry cross-laboratory

controls (NASA $\leftrightarrow$ CALCE and Severson $\rightarrow$ Oxford) that separate dataset shift from chemistry shift;

- an uncertainty treatment in which the cell, not the cycle, is the experimental unit: cell-level bootstrap confidence intervals on every headline number, with cycle-level DeLong tests demoted to secondary evidence and their pseudoreplication caveat stated explicitly;
- a reliability analysis of probability calibration under shift — Platt, isotonic, and temperature scaling, all fitted on out-of-fold source scores — including recalibration on small labeled LFP samples, and a cost-sensitive operational evaluation with source-selected decision thresholds;
- a complete embedded deployment that packs all three ensembles into one flat binary and runs them on an ESP32-S3 through a hand-written C tree walker.

The remainder of this paper is organized as follows. Section II describes the methodology, including the failure labels and their endpoints, the datasets, the models including the discrete-time hazard formulation, the calibration and transfer protocols, and the deployment architecture. Section III presents the experimental results. Section IV discusses the analysis, implications, and limitations, and Section V concludes.

II. Methodology

II-A. Composite Failure Label

Following the original protocol [7], the State-of-Health of cell c at cycle k is the ratio of the discharge capacity to the mean capacity over the first 10 cycles,

$$\begin{aligned}\text{SOH}_k^{(c)} &= \frac{Q_k^{(c)}}{\bar{Q}^{(c)}}, \\ \bar{Q}^{(c)} &= \frac{1}{10} \sum_{i=1}^{10} Q_i^{(c)},\end{aligned}\tag{1}$$

where $Q_k^{(c)}$ is the discharge capacity at cycle k . The early-life voltage baseline is defined analogously, $\bar{V}^{(c)} = \frac{1}{10} \sum_{i=1}^{10} \bar{V}_i^{(c)}$, with $\bar{V}_i^{(c)}$ the average discharge

voltage. A discharge cycle t in cell c is labeled as failing within horizon H if any cycle $k \in [t, t+H)$ meets either of two criteria,

$$y_t^{(c,H)} = \begin{cases} 1, & \exists k \in [t, t+H) : \text{SOH}_k^{(c)} \leq 0.80 \text{ or} \\ & \bar{V}_k^{(c)} < 0.94 \bar{V}^{(c)}, \\ 0, & \text{otherwise,} \end{cases} \quad (2)$$

and once triggered the label stays 1 for all later cycles. Because the window $[t, t+H)$ includes the current cycle, rows scored at or after the failure crossing are labeled positive for every formulation in this paper: with-SOH results therefore mix concurrent failure detection with future failure prediction, and Section III-C reports the two components separately. This paragraph makes explicit what was already true of every evaluation reported here; the label code is unchanged and no reported number moves as a result of the clarification. The task is to estimate the cumulative failure risk over the horizon,

$$p_t^{(c,H)} = \Pr(T_f \leq t+H \mid T_f > t, \mathbf{x}_t^{(c)}), \quad (3)$$

where T_f is the failure cycle and $\mathbf{x}_t^{(c)}$ is the feature vector measured at cycle t . Datasets without voltage data, or where voltage sits at the cutoff as in Oxford’s flat LFP plateau, fall back to the SOH criterion alone.

Three properties of this label definition matter for everything that follows. First, (3) is a fixed-horizon cumulative failure risk, not a discrete-time hazard: each horizon is an independent binary classifier, no cycle-level hazard h_k or survival product is estimated, and no monotonicity constraint $p_t^{(c,10)} \leq p_t^{(c,20)} \leq p_t^{(c,30)} \leq p_t^{(c,50)}$ is enforced. Section II-G adds a true discrete-time hazard model, and Section III-H quantifies how often the four independent classifiers violate monotonicity. Second, the label is *partially defined by SOH itself*, and SOH is also an input feature; a with-SOH model can therefore detect proximity to the $\text{SOH} = 0.80$ label threshold without learning any degradation dynamics. We treat every with-SOH result in this paper as an upper bound that includes this label-proximity leakage, and Section III-F measures it directly by relabeling failures with a single endpoint at a time. Third, the two endpoints can be separated: the label can be defined by the SOH criterion alone, by the voltage-sag criterion alone, or by both, and the model’s feature set can be ablated independently of the label definition.

II-B. Datasets

Eight public aging datasets are used:

- **1) NASA 18650** [18]: 37 LCO cells aged under randomized charge–discharge profiles at room temperature, 1 028 cleaned discharge rows in total (about 28 per cell), with diverse failure modes;
- **2) CALCE LCO/CX2** [19]: 7 cells (CS2_33–36, CX2_36–38) at 1C/1C with per-cycle voltage, current, and capacity; per-cell ambient temperature and discharge duration are unlogged;
- **3) Oxford** [20]: 5 LFP pouch cells at 1C/1C under controlled temperature. The public release stores discharge summaries at roughly 100-cycle intervals, so after cleaning, 46–78 usable discharge cycles per cell remain (320 rows total);
- **4) MIT–Stanford Severson** [21, 22]: 141 LFP cells under a fast-charging protocol, 101–2 237 cleaned cycles each (about 117 000 rows in total);
- **5) HNEI NMC/LCO** [23]: 15 NMC/LCO-blend 18650 cells cycled continuously at 25 °C under a fixed charge protocol and mixed discharge C-rates, 1016–1064 cycles each — all reaching the 80% SOH end-of-life criterion;
- **6) SNL NCA** [23, 24]: 18 NCA cells at 15–35 °C across several C-rate combinations, 451–904 cycles each;
- **7) SNL NMC** [23, 24]: 6 NMC 18650 cells at 15–25 °C across mixed C-rates, 384–670 cycles each;
- **8) SNL LFP** [23, 24]: 21 LFP cells cycled at 15–35 °C, 3022–4544 cycles each, of which 5 reach 80% SOH within the recorded window.

Items 5–8 are the Battery Archive “Cycle Test Data” dashboard export [23] (retrieved September 2026; per-cycle capacity and voltage summaries), whose underlying SNL test matrix is documented by Preger et al. [24]; the groups are used here only as *transfer targets* and for within-group controls; none of their cycles influences protocol or hyperparameter selection. The Battery Archive export provides per-cycle capacity and voltage summaries but, like CALCE, no per-cycle current or temperature columns, so the affected features are zero-imputed exactly like CALCE’s, and the unit-safe common feature set is primary for these groups.

A pre-registered cleaning rule, fixed before any model was trained on the new cells, admits a cell only if it passes: formation-cycle removal by low first-cycle coulombic efficiency, a Hampel despiking of the capacity trace, truncation at sustained level shifts in capacity (re-test or rest artifacts), per-cycle voltage validity, and a minimum of 20 usable cycles. Of 75 downloaded cells, 60 pass (15 + 6 NMC, 18 NCA, 21 LFP; 106 787 cycle rows); the remaining 15 are excluded for exactly one recorded reason each — in this download, all 15 for state-of-charge windows mixing partial and reference cycles — and are never used. On every admitted cell the SOH end-of-life crossing occurs no later than the voltage-sag crossing, so the composite failure label on the new groups is, in effect, the SOH criterion alone.

One consequence of the Oxford logging resolution deserves emphasis. Because consecutive cleaned cycles are roughly 100 cycles apart, every horizon $H \in \{10, 20, 30, 50\}$ is shorter than the sampling interval, and the set of positive-label rows on Oxford is *identical for all four horizons*; Oxford scores therefore measure pooled failure detection rather than horizon-resolved prediction, and AUC differences between horizons on Oxford come entirely from the source-side model, not from the target labels. Per-cell statistics on Oxford nominally cover all five cells, but one (Cell5) contributes a single positive row, so its within-cell AUC is uninformative and per-cell means are effectively four-cell statistics. Both caveats apply to every Oxford number in this paper.

II-C. Features and Preprocessing

Each cycle is described by seven scalar features: cycle index, average discharge voltage, minimum discharge voltage, average discharge current, average temperature, discharge duration, and SOH. Trees are scale-invariant, so no standardization is applied to them; the GRU and all logistic-regression baselines receive per-feature standardization using training-set statistics. Missing CALCE values (temperature and duration are unlogged) are filled with zero; the Battery Archive groups lack the same columns and are zero-imputed identically, so four of the eight datasets contain at least one structurally missing feature column. Within a single dataset a constant feature cannot be used for splitting, but in pooled cross-dataset training a constant zero column acts as a dataset identifier; Section III-G quantifies this confound with a common-feature control.

II-D. Tree Models and Hyperparameters

Three tree ensembles are benchmarked with hyperparameters matched to the original study [7]: XGBoost [5] with max_depth 4, 300 estimators, learning rate 0.05, subsample 0.8, colsample_bytree 0.8, and min_child_weight 5; LightGBM [6] with the same depth, count, and rate; and Random Forest [4] with max_depth 6 and 300 estimators. All use random_state 42 for deterministic replication.

II-E. GRU Sequence Classifier

A compact GRU tests whether sequence-aware representations capture trajectories that transfer across chemistries. It ingests a sliding window of $W=10$ consecutive cycles per cell as a $10 \times f$ tensor and runs one GRU layer with 8 hidden units,

$$\begin{aligned} \mathbf{r}_t &= \sigma(W_r \mathbf{x}_t + U_r \mathbf{h}_{t-1}), \\ \mathbf{z}_t &= \sigma(W_z \mathbf{x}_t + U_z \mathbf{h}_{t-1}), \\ \tilde{\mathbf{h}}_t &= \tanh(W_h \mathbf{x}_t + U_h (\mathbf{r}_t \odot \mathbf{h}_{t-1})), \\ \mathbf{h}_t &= (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t, \end{aligned} \tag{4}$$

and the final hidden state maps through a linear layer with sigmoid activation, $p = \sigma(\mathbf{w}^\top \mathbf{h}_W + b)$, to give the failure probability. Adam with learning rate 0.005 and binary cross-entropy loss are used. The 8-unit width was chosen to match the small number of training cells (7 to 37); wider layers overfit severely. For cross-dataset runs the feature set is reduced to the common, unit-safe set — cycle number, average voltage, minimum voltage, and SOH when enabled — standardized with training-set statistics; average current is dropped (amperes for LCO sources, derivative-based estimates for the LFP targets), as are temperature and duration. The trees are additionally evaluated on this identical common feature set (Section III-C), so that architecture and feature availability are not confounded: where the GRU and the trees use different inputs, we say so explicitly, and no architecture claim rests on a feature-confounded comparison.

II-F. Minimal Baselines

Because the central question is whether apparent performance is a health-state shortcut, the most important models are embarrassingly simple ones [7]:

- **SOH distance rule:** the fitted-free score $d_t = 0.80 - \text{SOH}_t$, i.e., distance below the label threshold;
- **SOH-only** logistic regression, $\Pr(y_t^{(H)}=1) = \sigma(a_H \text{SOH}_t + b_H)$;
- **cycle-only** logistic regression on the cycle index;
- **SOH + cycle** logistic regression;
- **sensors-only** logistic regression on the five non-SOH, non-cycle features;
- **full-feature** logistic regression on all seven features.

If the SOH-only and distance-rule baselines reproduce most of the transferable performance of the tuned ensembles, then the ensembles contribute little beyond reading the health state — which is precisely the negative result the validity question predicts, and the one an evaluation-oriented paper should surface.

II-G. A True Discrete-Time Hazard Model

To complement the four independent fixed-horizon classifiers, we also fit an explicit discrete-time survival model. For each observed cycle t and each offset $\ell \in \{1, \dots, L\}$ with $L=50$, define the hazard

$$h_\ell(t) = \Pr(T_f = t+\ell \mid T_f \geq t+\ell, \mathbf{x}_t), \quad (5)$$

conditional on the features measured at t only, so the formulation remains operational (no future measurements are required). One pooled classifier — XGBoost with the same hyperparameters as above, and a logistic regression — is fit on the offset-expanded at-risk training rows $(\mathbf{x}_t, \ell)$ with target $\mathbf{1}\{T_f = t+\ell\}$, restricted to rows with $t+\ell \leq T_f$, and with ℓ as an additional input feature. The cumulative risk over horizon H is the survival product

$$R(t, H) = 1 - \prod_{\ell=1}^H (1 - \hat{h}_\ell(t)), \quad (6)$$

which is monotone non-decreasing in H by construction and yields all four horizons from one model. Rows at or after failure are scored exactly like the fixed-horizon classifiers score them, so the evaluation rows are identical across formulations.

II-H. Calibration Protocol

Three post-hoc calibration schemes are compared: Platt scaling fits a sigmoid to the base model’s raw scores s_i ,

$$p_i = \frac{1}{1 + \exp(-\alpha s_i - \beta)}, \quad (7)$$

by logistic regression; isotonic regression [11] fits the non-decreasing step function that minimizes squared error, implemented with scikit-learn’s `IsotonicRegression` and out-of-bounds clipping; and temperature scaling divides the logit of the score by a single fitted temperature.

The protocol is leakage-clean. For every within-dataset evaluation fold, the calibrators are fitted on *cross-fitted out-of-fold scores*: an inner cell-disjoint GroupKFold over the training cells produces, for every training row, a score from a model that never saw that row’s cell, and the calibrators are fitted on those out-of-fold scores; the untouched test cells are scored by a model trained on all training cells and calibrated with the out-of-fold-fitted functions. In the transfer setting the transfer model trains on all source cells, the calibrators are fitted on source out-of-fold scores, and the target cells are scored and calibrated without ever entering any fit. No calibrator in this paper is ever fitted on the scores it is applied to, and no test cell contributes to training or calibration. The earlier version of this study fitted calibrators on in-sample training scores; the cross-fitted absolute calibration numbers reported here are accordingly more conservative.

A reproducibility note is retained from the original study comparison. The original study [7] reported Brier scores around 0.032; re-execution of the released code yields Brier values between roughly 0.09 and 0.29, a gap of up to ninefold that could not be resolved from the available source. The AUC values reported here are comparable to the published range, and the qualitative conclusions hold. The Brier gap does not affect the relative comparisons that form the core of this paper.

II-I. Transfer Protocols

Transfer is evaluated by training on LCO cells (NASA alone, CALCE alone, or both) and testing on the cells of the target datasets. Performance is reported per cell (mean $\pm$ standard deviation across test cells) and pooled over all target rows with cell-level bootstrap confidence intervals; pooling alone can hide degenerate behavior where a model assigns nearly identical

scores to every cell, and per-cell statistics expose it. Three families of transfer are used:

- **Cross-chemistry** (LCO→LFP): NASA, CALCE, or both → Oxford and → Severson, with and without SOH as a feature, and on the common feature set shared with the GRU;
- **Same-chemistry, cross-laboratory controls**: NASA→CALCE and CALCE→NASA (LCO→LCO) and Severson→Oxford and Oxford→Severson (LFP→LFP). Because chemistry, laboratory, cell manufacturer, cycling protocol, temperature, and logging resolution all change simultaneously between any source and target, the cross-chemistry shifts above are cross-*dataset* shifts as well; “cross-chemistry” is used throughout as shorthand, and these controls are what allow dataset shift and chemistry shift to be told apart;
- **Feature-definition ablations**: the factorial feature grid of Section III-E and the failure-endpoint relabeling of Section III-F.

II-J. Evaluation and Statistical Testing

Within-dataset evaluation uses 5-fold cell-grouped cross-validation on NASA (37 cells) and leave-one-cell-out on CALCE (7 cells), so every cycle from a given cell stays entirely on one side of every split — train, calibration, and test are cell-disjoint by construction, and models are retrained from scratch per horizon. The experimental unit is the cell, not the cycle: cycles within a cell are strongly correlated, so headline AUCs are reported as fold means with fold standard deviations, and every headline number carries a 95% bootstrap confidence interval obtained by resampling *cells* with replacement (2 000 draws, or 400 for the 117 000-row Severson pools) and recomputing the statistic. Paired comparisons — above all with-SOH versus without-SOH — use the same cell resamples for both models, giving a paired bootstrap interval on the AUC difference as the primary evidence.

The DeLong nonparametric test [25] for paired ROC curves is retained as secondary evidence. It evaluates

$$z = \frac{A_1 - A_2}{\sqrt{V_{11} + V_{22} - 2V_{12}}}, \quad (8)$$

where A_j is the empirical area under the j -th ROC curve and V_{ij} are covariance terms estimated from the same test instances. The test treats every

cycle-level prediction as an independent observation; because cycles within a cell are correlated, the effective sample size is far smaller than the row count, and the resulting p -values are inflated by pseudoreplication. We therefore report them with that caveat and read them only as evidence that an effect is directionally consistent, never as precise significance statements.

Discrimination alone is not enough for a reliability-oriented study, so each within-dataset configuration also reports the Brier score, expected calibration error (10 equal-width bins), log loss, the slope and intercept of a logistic recalibration of the predicted probabilities (slope 1 and intercept 0 indicate perfect calibration), the area under the precision–recall curve, which is sensitive to the horizon-dependent class prevalence that AUC ignores, and sensitivity at a fixed 10% false-positive rate. Operational, cost-sensitive evaluation is described with the results in Section III-K.

Three reporting conventions apply throughout. First, pooled AUC aggregates rows across cells, so long-lived cells dominate it; per-cell mean $\pm$ SD and cell-level bootstrap intervals are the primary evidence, and pooled numbers are never read without them. Second, failure prevalence rises with the horizon, so Brier score, log loss, calibration error, decision thresholds, and precision–recall performance are horizon-dependent quantities that cannot be compared across H without that adjustment in mind. Third, the operational cost model prices a missed failure at 20 times a false alarm; decision thresholds are selected on source out-of-fold scores under a 10% false-alarm budget and evaluated under that 20:1 expected cost on the target. Sensitivity of the conclusions to other budgets and ratios is not tested and is stated as a limitation.

II-K. Embedded Deployment Architecture

The deployment pipeline exports the three trained models into one flat binary and runs them on an ESP32-S3 through a hand-written C tree walker. The layout is deliberately simple: a 12-byte file header, a 3×4 -byte offset table, and per-model sections holding a 4-byte tree count followed by consecutive per-tree node arrays. Each node stores the split feature index, the split threshold as a 32-bit float, and child indices; leaves store the class-1 probability directly. The walker is one C function of about 70 lines that follows child pointers, sums the leaf contributions per tree, and applies the per-model output transform. Writing ℓ_t for the leaf value reached in tree t and T for the number of trees, Random Forest averages and clamps its leaf

probabilities,

$$p = \text{clamp}\left(\frac{1}{T} \sum_{t=1}^T \ell_t, 0, 1\right), \quad (9)$$

while XGBoost and LightGBM pass the summed score through a sigmoid,

$$p = \frac{1}{1 + \exp(-\sum_{t=1}^T \ell_t - s_0)}, \quad (10)$$

where s_0 is the model’s initialization score ($s_0 \neq 0$ only for XGBoost). Three validation stages confirm correctness: a Python walker that mirrors the C logic and is compared against `predict_proba()`, a cross-compiled C binary tested on desktop x86_64, and on-target execution on an ESP32-S3-DevKitC-1 reading the binary from flash. All three must report maximum absolute error below 10^{-5} on every row before deployment is accepted.

III. Results

III-A. Within-Dataset Performance

Table 2 summarizes within-dataset performance across the four horizons, reported as fold-mean AUC $\pm$ fold standard deviation under the leakage-clean protocol (leave-one-cell-out on CALCE, 5-fold cell-grouped cross-validation on NASA), together with the Platt Brier score. Discrimination is strong throughout: every tree configuration clears a fold-mean AUC of 0.89, on both datasets, at every horizon (Fig. 1). The GRU reaches comparable means but with visibly higher fold-to-fold variance, and the discrete-time hazard model (Section II-G) lands in the same range as the fixed-horizon XGBoost classifier while covering all horizons with a single fit. On the four new Battery Archive groups (Section II-B), under the same protocol and per-group inner-knowledge settings of Section II-C, discrimination remains strong: 38 of the 39 tree-group-horizon configurations reach a fold-mean AUC of at least 0.85, spanning 0.61–1.00 across the NMC, NCA and LFP groups — although the LFP-SNL group’s 0.031 positive-label prevalence and mostly censored cells make its numbers supporting rather than headline evidence.

III-B. Calibration Under Cross-Fitting

Table 3 compares the three calibrators, averaged over the tree models and horizons, where every calibrator is fitted on cross-fitted out-of-fold scores

Table 2: Within-Dataset Failure-Risk Performance Across Four Horizons (Fold-Mean AUC $\pm$ Fold SD; Tree Ensembles and GRU Platt-Calibrated, Hazard Models Raw; CALCE Uses Leave-One-Cell-Out, NASA 5-Fold Cell-Grouped CV). Brier Is the Platt Pooled Value Averaged Over Horizons (Raw for Hazard Rows).

Model	Dataset	H=10	H=20	H=30	H=50	Mean AUC	Brier
XGBoost	NASA	0.903 $\pm$ 0.097	0.899 $\pm$ 0.121	0.910 $\pm$ 0.111	0.918 $\pm$ 0.115	0.907	0.182
LightGBM	NASA	0.912 $\pm$ 0.065	0.909 $\pm$ 0.126	0.892 $\pm$ 0.121	0.901 $\pm$ 0.073	0.903	0.206
Random Forest	NASA	0.911 $\pm$ 0.106	0.901 $\pm$ 0.109	0.912 $\pm$ 0.113	0.927 $\pm$ 0.107	0.913	0.201
GRU	NASA	0.872 $\pm$ 0.122	0.893 $\pm$ 0.114	0.884 $\pm$ 0.083	0.856 $\pm$ 0.128	0.876	0.307
Hazard XGBoost	NASA	0.918 $\pm$ 0.086	0.935 $\pm$ 0.086	0.937 $\pm$ 0.081	0.927 $\pm$ 0.070	0.929	0.182
Hazard Logistic	NASA	0.902 $\pm$ 0.064	0.899 $\pm$ 0.057	0.896 $\pm$ 0.045	0.867 $\pm$ 0.052	0.891	0.175
XGBoost	CALCE	0.965 $\pm$ 0.041	0.967 $\pm$ 0.036	0.967 $\pm$ 0.040	0.943 $\pm$ 0.071	0.961	0.075
LightGBM	CALCE	0.967 $\pm$ 0.047	0.968 $\pm$ 0.038	0.943 $\pm$ 0.070	0.942 $\pm$ 0.076	0.955	0.075
Random Forest	CALCE	0.934 $\pm$ 0.074	0.933 $\pm$ 0.070	0.917 $\pm$ 0.098	0.903 $\pm$ 0.110	0.922	0.070
GRU	CALCE	0.952 $\pm$ 0.072	0.959 $\pm$ 0.083	0.935 $\pm$ 0.090	0.921 $\pm$ 0.090	0.942	0.130
Hazard XGBoost	CALCE	0.894 $\pm$ 0.135	0.893 $\pm$ 0.130	0.916 $\pm$ 0.095	0.889 $\pm$ 0.096	0.898	0.108
Hazard Logistic	CALCE	0.979 $\pm$ 0.042	0.948 $\pm$ 0.119	0.936 $\pm$ 0.138	0.945 $\pm$ 0.117	0.952	0.122

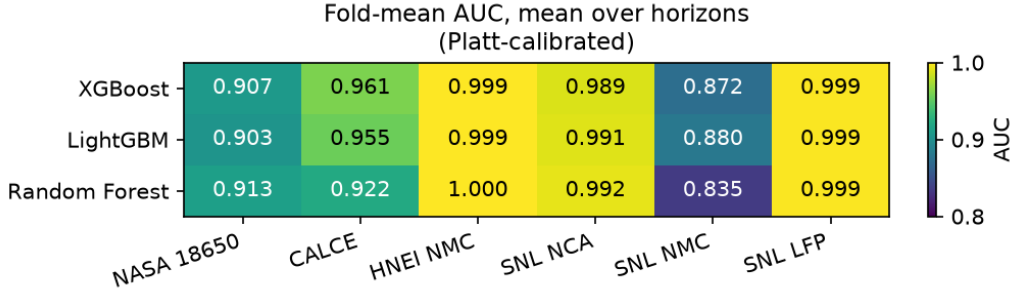


Figure 1: Within-dataset AUC heatmap for the tree ensembles, mean across horizons, Platt-calibrated. Right block: the four new Battery Archive groups. GRU and hazard within-dataset numbers are in Table 2.

(Section II-H). Three observations follow. First, on discrimination the raw scores are effectively unbeatable within dataset: any fitted calibrator is a monotone re-mapping and preserves per-fold ranking up to the ties it introduces, and the fold-mean AUC differences between calibrators are correspondingly small. Second, the step function of isotonic regression creates large tied groups whose cost is invisible to fold-mean ROC-AUC but severe for precision-recall: on NASA, raw scores reach a mean PR-AUC of 0.774 while isotonic collapses to 0.639 (Fig. 3). Third, calibration quality proper is best for Platt on CALCE (ECE 0.029 against 0.049 for isotonic), but the fitted Platt slope on CALCE is only 0.011: on that long-tailed, leave-one-cell-out setting the out-of-fold sigmoid essentially washes confidence out toward

Table 3: Calibration Quality Averaged Over Tree Models and Horizons. All Calibrators Are Fitted on Cross-Fitted Out-of-Fold Scores. Slope 1 / Intercept 0 Indicate Perfect Probabilistic Calibration.

Dataset	Method	AUC (fold mean)	Brier	ECE	Log loss	Slope	Intercept
NASA	Isotonic	0.854	0.207	0.151	0.652	0.48	-0.26
NASA	Platt	0.908	0.196	0.136	0.575	0.90	-0.14
NASA	Temperature	0.908	0.178	0.107	0.536	1.22	-0.30
CALCE	Isotonic	0.915	0.067	0.049	0.279	0.41	1.17
CALCE	Platt	0.946	0.073	0.029	0.280	0.01	2.54
CALCE	Temperature	0.945	0.080	0.067	0.284	0.53	1.42

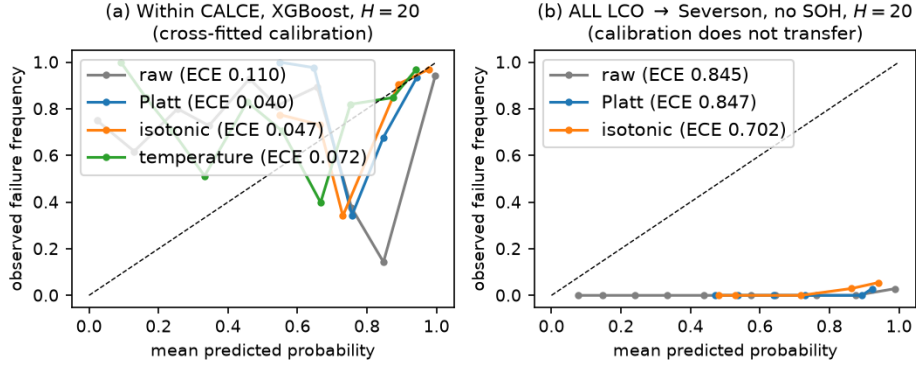


Figure 2: Reliability diagrams under the cross-fitted protocol. (a) Within CALCE, XGBoost at $H=20$: the fitted Platt sigmoid preserves ranking but washes confidence toward the base rate. (b) ALL-LCO → Severson without SOH at $H=20$: no fitted calibrator, whichever source scores it is applied to, repairs calibration on the shifted target.

the base rate while preserving ranking. Temperature scaling is the most balanced compromise on NASA (slope 1.218, Brier 0.178, both better than Platt’s). There is no global winner across datasets: temperature leads on NASA, Platt on CALCE ECE at the cost of a destroyed slope. We therefore read the comparison as: *Platt preserves discrimination best while achieving calibration error comparable to or better than isotonic*, which is a materially weaker and more defensible claim than “Platt is the better calibrator,” and raw scores remain the safest ranking signal whenever the operating context is uncertain.

III-C. Cross-Dataset Transfer With and Without SOH

Table 4 reports LCO→LFP transfer at $H=20$ with SOH kept, and Table 5 the same experiments without SOH; Fig. 4 summarizes both as a map

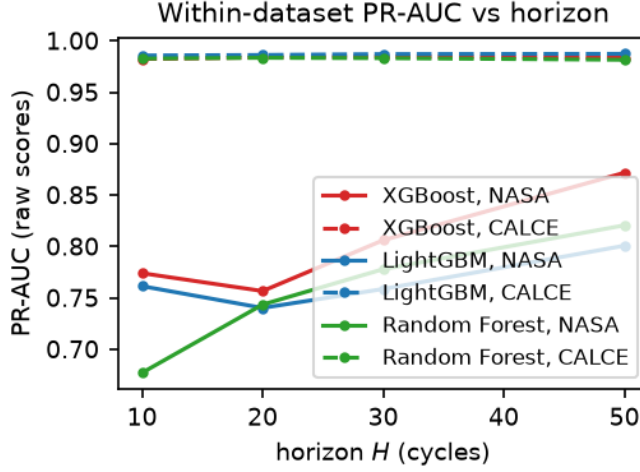


Figure 3: Within-dataset PR-AUC of raw scores versus horizon. Precision–recall performance, unlike ROC-AUC, is sensitive to the horizon-dependent failure prevalence; it is also where isotonic’s tied step outputs pay for their ranking-free compression.

of the AUC drop caused by removing SOH. Every number carries a cell-level bootstrap 95% confidence interval on the pooled rows, in addition to the per-cell mean and standard deviation. With SOH available, the tree ensembles reach pooled raw AUC of up to 1.00 (Oxford) and 0.90 (Severson). Removing SOH collapses transfer to at most 0.58 pooled on Oxford, with Severson between 0.53 and 0.75 — point estimates fall below 0.5 in several configurations, significantly so for the discrete-time hazard model on Oxford, while elsewhere the intervals still cover chance and the safe reading is lost signal with occasional inversion.

The GRU behaves no better on aggregate, and its multi-seed evaluation (three seeds) shows a source-dependent instability: seed-averaged per-cell AUC on Severson without SOH is 0.987 ± 0.015 for the NASA source (stable, near-oracle) but 0.479 ± 0.451 for CALCE (seeds spanning 0.21 to 1.00) and 0.893 ± 0.131 for the pooled source — against 0.998 ± 0.003 , 0.964 ± 0.033 and 0.991 ± 0.015 with SOH. The instability is therefore a property of particular source–target pairings rather than uniform across training sources. On the identical common feature set the trees retain their table 4 behavior, so the GRU–tree gap under shift cannot be attributed to the GRU’s reduced inputs: the sequence model’s distributed hidden state genuinely offers no cross-dataset generalization advantage here. On Oxford the GRU is at or

Table 4: Cross-Dataset Transfer at H=20 With SOH (Upper-Bound / Label-Proximity Regime). Raw-Score Pooled AUC With Cell-Level Bootstrap 95% CI; Per-Cell Mean $\pm$ SD Across Test Cells (Oxford: Five Cells, One With a Single Positive Row; See Section II-B).

Training	Model	Oxford		Severson	
		Pooled AUC [CI]	Per-cell	Pooled AUC [CI]	Per-cell
NASA	XGBoost	0.957 [0.931, 0.987]	0.959 $\pm$ 0.037	0.871 [0.767, 0.997]	0.997 $\pm$ 0.010
NASA	LightGBM	1.000 [1.000, 1.000]	1.000 $\pm$ 0.000	0.849 [0.724, 0.998]	0.992 $\pm$ 0.048
NASA	Random Forest	1.000 [1.000, 1.000]	1.000 $\pm$ 0.000	0.861 [0.739, 0.998]	0.999 $\pm$ 0.007
CALCE	XGBoost	0.850 [0.764, 0.910]	0.763 $\pm$ 0.245	0.850 [0.749, 0.967]	0.957 $\pm$ 0.054
CALCE	LightGBM	0.889 [0.827, 0.932]	0.837 $\pm$ 0.193	0.863 [0.782, 0.957]	0.952 $\pm$ 0.037
CALCE	Random Forest	0.850 [0.773, 0.899]	0.766 $\pm$ 0.169	0.902 [0.834, 0.984]	0.996 $\pm$ 0.022
ALL LCO	XGBoost	0.917 [0.878, 0.965]	0.826 $\pm$ 0.265	0.896 [0.812, 0.995]	0.995 $\pm$ 0.015
ALL LCO	LightGBM	0.971 [0.959, 0.981]	0.969 $\pm$ 0.015	0.882 [0.789, 0.990]	0.993 $\pm$ 0.026
ALL LCO	Random Forest	0.989 [0.979, 0.998]	0.992 $\pm$ 0.013	0.889 [0.796, 0.998]	0.996 $\pm$ 0.021

Table 5: Cross-Dataset Transfer at H=20 Without SOH (Prognostically Meaningful Regime). Columns as in Table 4.

Training	Model	Oxford		Severson	
		Pooled AUC [CI]	Per-cell	Pooled AUC [CI]	Per-cell
NASA	XGBoost	0.518 [0.513, 0.522]	0.519 $\pm$ 0.006	0.595 [0.510, 0.726]	0.676 $\pm$ 0.184
NASA	LightGBM	0.533 [0.497, 0.543]	0.448 $\pm$ 0.201	0.527 [0.422, 0.672]	0.535 $\pm$ 0.183
NASA	Random Forest	0.521 [0.487, 0.537]	0.442 $\pm$ 0.198	0.674 [0.561, 0.826]	0.836 $\pm$ 0.176
CALCE	XGBoost	0.416 [0.264, 0.554]	0.360 $\pm$ 0.208	0.682 [0.670, 0.695]	0.806 $\pm$ 0.082
CALCE	LightGBM	0.462 [0.307, 0.579]	0.410 $\pm$ 0.207	0.643 [0.530, 0.721]	0.830 $\pm$ 0.042
CALCE	Random Forest	0.501 [0.320, 0.641]	0.435 $\pm$ 0.243	0.665 [0.547, 0.749]	0.870 $\pm$ 0.049
ALL LCO	XGBoost	0.429 [0.279, 0.597]	0.415 $\pm$ 0.240	0.750 [0.692, 0.835]	0.860 $\pm$ 0.139
ALL LCO	LightGBM	0.332 [0.112, 0.515]	0.316 $\pm$ 0.209	0.718 [0.634, 0.850]	0.808 $\pm$ 0.167
ALL LCO	Random Forest	0.581 [0.516, 0.644]	0.608 $\pm$ 0.103	0.746 [0.673, 0.845]	0.876 $\pm$ 0.115

below chance (Table 6), far worse than the tree ensembles on the same target.

The reading that survives the uncertainty treatment is deliberately modest. SOH appears on both sides of the learning problem — in the feature vector and inside the label definition — so high with-SOH transfer is compatible with two mechanisms: a chemistry-dependent mapping from SOH to remaining life, and mere detection of proximity to the SOH = 0.80 criterion that defines the label. The experiments that follow separate these mechanisms.

Diagnosis or prognosis? Because the label window includes the scoring cycle, rows from already-failed cells score positive. Table 7 splits every head-line transfer number into all-rows AUC and prognosis-only AUC computed

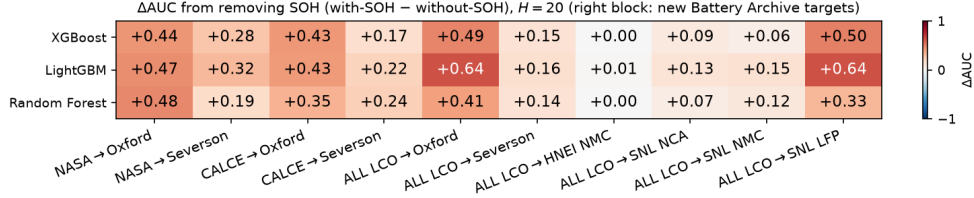


Figure 4: Transferability collapse map at $H=20$ for the tree ensembles: the AUC drop from removing SOH, per source→target pair and model. Positive values mean SOH removal destroys apparent transfer performance; the near-uniform saturation at the maximum drop is the signature of a single-feature shortcut rather than of distributed degradation knowledge. Right block: ALL-LCO → new Battery Archive targets. GRU deltas are in the text and Table 6.

Table 6: GRU Transfer at $H=20$ on the Common Feature Set: Seed-Averaged Per-Cell AUC ($\pm$ SD Over Seeds 42, 1, 7).

Training	Feature set	Oxford	Severson
NASA	common w/ SOH	0.196 $\pm$ 0.155	0.998 $\pm$ 0.003
NASA	common no SOH	0.147 $\pm$ 0.210	0.987 $\pm$ 0.015
CALCE	common w/ SOH	0.182 $\pm$ 0.311	0.964 $\pm$ 0.033
CALCE	common no SOH	0.099 $\pm$ 0.171	0.479 $\pm$ 0.451
ALL LCO	common w/ SOH	0.088 $\pm$ 0.084	0.991 $\pm$ 0.015
ALL LCO	common no SOH	0.000 $\pm$ 0.000	0.893 $\pm$ 0.131

on the same scores restricted to rows scored while SOH is still above 0.80 — no retraining, only restricted rows. On Severson at $H=20$, XGBoost with SOH moves from 0.896 to 0.838 [0.760, 0.991] and without SOH from 0.750 to 0.721 [0.666, 0.833]: 99% of Severson rows are scored above threshold, so concurrent detection explains only about 0.05–0.07 of the with-SOH transfer there, and the remainder is near-threshold ranking — the proximity component that the failure-definition ablation prices in Section III-F. The distance rule behaves identically (0.912 to 0.862), as a pure threshold proxy must. On the new targets the diagnostic share is larger wherever cells linger below threshold — HNEI 0.993 to 0.963 (28

Table 7: Diagnosis Versus Prognosis at H=20 (ALL-LCO Source, Raw Scores): All-Rows Pooled AUC [Cell-Level CI Where Computed] Against Prognosis-Only AUC on Rows Scored While SOH>0.80, With the Share of Rows Scored Above Threshold. GRU and Distance-Rule Rows Are Point Estimates.

Target	Model	Features	All rows	Prognosis-only	Frac. above
Severson	GRU	common, no SOH	0.631	0.553	0.99
Severson	GRU	common+SOH	0.827	0.756	0.99
Severson	LightGBM	no SOH	0.718 [0.634,0.846]	0.674 [0.597,0.851]	0.99
Severson	LightGBM	with SOH	0.882 [0.786,0.990]	0.814 [0.732,0.978]	0.99
Severson	SOH-dist rule	dist rule	0.912	0.862	0.99
Severson	SOH-dist rule	SOH-only	0.912	0.862	0.99
Severson	Random Forest	no SOH	0.746 [0.673,0.845]	0.705 [0.632,0.843]	0.99
Severson	Random Forest	with SOH	0.889 [0.783,0.998]	0.825 [0.737,0.996]	0.99
Severson	XGBoost	no SOH	0.750 [0.692,0.835]	0.721 [0.666,0.833]	0.99
Severson	XGBoost	with SOH	0.896 [0.804,0.995]	0.838 [0.760,0.991]	0.99
HNEI NMC	LightGBM	no SOH	0.988 [0.974,0.996]	0.905 [0.863,0.944]	0.28
HNEI NMC	LightGBM	with SOH	0.999 [0.998,0.999]	0.969 [0.946,0.987]	0.28
HNEI NMC	Random Forest	no SOH	0.996 [0.994,0.999]	0.945 [0.908,0.978]	0.28
HNEI NMC	Random Forest	with SOH	0.999 [0.999,1.000]	0.981 [0.970,0.993]	0.28
HNEI NMC	XGBoost	no SOH	0.989 [0.977,0.997]	0.908 [0.870,0.945]	0.28
HNEI NMC	XGBoost	with SOH	0.993 [0.983,0.999]	0.963 [0.935,0.987]	0.28
SNL NCA	LightGBM	no SOH	0.850 [0.791,0.911]	0.841 [0.798,0.889]	0.63
SNL NCA	LightGBM	with SOH	0.985 [0.974,0.993]	0.891 [0.845,0.946]	0.63
SNL NCA	Random Forest	no SOH	0.911 [0.875,0.951]	0.837 [0.779,0.907]	0.63
SNL NCA	Random Forest	with SOH	0.983 [0.974,0.992]	0.901 [0.858,0.957]	0.63
SNL NCA	XGBoost	no SOH	0.878 [0.824,0.930]	0.849 [0.803,0.895]	0.63
SNL NCA	XGBoost	with SOH	0.972 [0.956,0.988]	0.886 [0.837,0.943]	0.63
SNL NMC	LightGBM	no SOH	0.693 [0.570,0.959]	0.728 [0.577,0.956]	0.60
SNL NMC	LightGBM	with SOH	0.847 [0.743,0.982]	0.722 [0.604,0.956]	0.60
SNL NMC	Random Forest	no SOH	0.742 [0.689,0.874]	0.649 [0.604,0.793]	0.60
SNL NMC	Random Forest	with SOH	0.858 [0.748,0.975]	0.706 [0.613,0.929]	0.60
SNL NMC	XGBoost	no SOH	0.709 [0.612,0.953]	0.730 [0.599,0.946]	0.60
SNL NMC	XGBoost	with SOH	0.772 [0.677,0.979]	0.729 [0.608,0.960]	0.60
SNL LFP	LightGBM	no SOH	0.359 [0.282,0.443]	0.383 [0.311,0.467]	0.97
SNL LFP	LightGBM	with SOH	0.999 [0.998,1.000]	0.993 [0.988,0.998]	0.97
SNL LFP	Random Forest	no SOH	0.666 [0.569,0.769]	0.588 [0.565,0.616]	0.97
SNL LFP	Random Forest	with SOH	0.999 [0.998,1.000]	0.994 [0.990,0.999]	0.97
SNL LFP	XGBoost	no SOH	0.496 [0.455,0.531]	0.517 [0.481,0.575]	0.97
SNL LFP	XGBoost	with SOH	0.998 [0.997,1.000]	0.991 [0.985,0.997]	0.97

III-D. Minimal Baselines: How Much Is the SOH Shortcut?

Table 8 places the tuned ensembles next to trivial baselines. The distance-to-threshold rule $d_t = 0.80 - \text{SOH}_t$, which involves no fitting at all, transfers to Severson at pooled AUC 0.912 [CI] for the ALL-LCO source — matching or exceeding the tuned ensembles’ with-SOH transfer in Table 4. An SOH-only logistic regression behaves equivalently (0.912). Both are diagnostic baselines by construction: they order cells by proximity to the label

Table 8: Minimal Transfer Baselines at $H=20$: Pooled AUC With Cell-Level Bootstrap 95% CI. The Distance Rule Is the Unfitted Score 0.80–SOH.

Baseline	Oxford			Severson		
	NASA src	CALCE src	ALL src	NASA src	CALCE src	ALL src
SOH distance rule	1.000 [1.000,1.000]	1.000 [1.000,1.000]	1.000 [1.000,1.000]	0.912 [0.836,0.998]	0.912 [0.836,0.998]	0.912 [0.836,0.998]
SOH only	1.000 [1.000,1.000]	1.000 [1.000,1.000]	1.000 [1.000,1.000]	0.912 [0.836,0.998]	0.912 [0.836,0.998]	0.912 [0.836,0.998]
cycle only	0.989 [0.983,1.000]	0.756 [0.732,0.780]	0.787 [0.760,0.814]	0.713 [0.591,0.798]	0.713 [0.591,0.798]	0.713 [0.591,0.798]
SOH + cycle	0.990 [0.985,1.000]	0.797 [0.770,0.825]	0.818 [0.788,0.848]	0.865 [0.834,0.902]	0.764 [0.674,0.829]	0.758 [0.664,0.825]
sensors only	0.500 [0.500,0.500]	0.500 [0.500,0.500]	0.500 [0.500,0.500]	0.301 [0.015,0.549]	0.498 [0.497,0.500]	0.303 [0.059,0.497]
full (7 feats)	0.500 [0.500,0.500]	0.500 [0.500,0.500]	0.500 [0.500,0.500]	0.823 [0.675,0.995]	0.712 [0.589,0.797]	0.674 [0.525,0.782]

threshold, not by learned degradation dynamics. By contrast, the sensor-only logistic regression transfers at 0.301 on Severson (NASA source): below chance, meaning the remaining sensor features actively invert on the shifted distribution. Within CALCE, the cycle index alone ranks failure at pooled AUC 0.936, approaching the tuned ensembles, because CALCE’s seven cells span very different cycle lives; on NASA, whose cells age under randomized profiles, the same feature is inverted (0.354).

Three conclusions follow. First, the apparent cross-dataset transferability of the tuned models is, to within the honest uncertainty, reproducible by a one-parameter health-threshold proxy: most of what transfers is the SOH coordinate, not degradation knowledge. Second, “more features” does not help transfer; the sensor features carry dataset-specific artifacts that can invert. Third, this outcome does not diminish the ensembles’ within-dataset value — there they beat every minimal baseline — but it does mean the transfer claim must be downgraded from “models transfer” to “the health state transfers, and models can read it.”

III-E. Factorial Feature Ablation

Table 9 runs the full $2^3 - 1$ grid over {SOH, cycle index, sensors} at $H=20$. Three regularities stand out. SOH-only is the strongest single group on both LFP targets, confirming that the health coordinate dominates whatever transfer exists. Cycle-only is competitive on Oxford — whose five cells fail at well-separated raw cycle counts, so the cycle index is a between-cell aging proxy — but the pattern is dataset-specific and does not survive to Severson. No combination without SOH reaches 0.76 on either target (best 0.75), and removing both SOH and the cycle index leaves the sensors with essentially nothing (0.535 with the ALL-LCO source on Severson). The ablation makes the shortcut diagnosis feature-by-feature: it is not that the models fail to use the sensor features; it is that the sensor features themselves carry no

Table 9: Factorial Feature Ablation at H=20 (XGBoost, Pooled AUC With Cell-Level Bootstrap 95% CI). Feature Groups: SOH; Cycle Index; Sensors = Voltages, Current, Temperature, Duration.

Feature set	Oxford			Severson		
	NASA src	CALCE src	ALL src	NASA src	CALCE src	ALL src
full	0.957 [0.931,0.987]	0.850 [0.764,0.910]	0.917 [0.878,0.965]	0.871 [0.767,0.997]	0.850 [0.749,0.967]	0.896 [0.812,0.995]
no SOH	0.518 [0.513,0.522]	0.416 [0.264,0.554]	0.429 [0.279,0.597]	0.595 [0.510,0.726]	0.682 [0.670,0.695]	0.750 [0.692,0.835]
—cycle	0.965 [0.942,0.989]	0.967 [0.929,0.981]	0.993 [0.954,1.000]	0.854 [0.733,0.998]	0.766 [0.584,0.974]	0.774 [0.578,0.998]
—SOH—cycle	0.658 [0.522,0.797]	0.433 [0.282,0.594]	0.388 [0.224,0.506]	0.531 [0.433,0.680]	0.500 [0.500,0.500]	0.535 [0.455,0.653]
SOH only	0.965 [0.942,0.989]	1.000 [1.000,1.000]	1.000 [1.000,1.000]	0.834 [0.695,0.991]	0.798 [0.638,0.978]	0.833 [0.689,0.993]
cycle only	0.541 [0.537,0.545]	0.541 [0.537,0.545]	0.541 [0.537,0.545]	0.709 [0.695,0.724]	0.694 [0.681,0.708]	0.719 [0.704,0.735]
sensors	0.658 [0.522,0.797]	0.433 [0.282,0.594]	0.388 [0.224,0.506]	0.531 [0.433,0.680]	0.500 [0.500,0.500]	0.535 [0.455,0.653]

cross-dataset signal.

III-F. Failure-Definition Ablation: Measuring the Label Circularity

Table 10 relabels failure by a single endpoint at a time — SOH $\leq$ 0.80 only, voltage sag only, or both — and repeats the transfer experiments with and without SOH as a feature. This is the experiment that quantifies the circularity concern of Section II-A, and its signature could hardly be cleaner: with-SOH transfer *tracks how much SOH defines the label*. When SOH alone defines failure — the model is, in effect, reading the label criterion — transfer to Severson is 0.985, essentially saturating. Under the combined label it is 0.896. Relabel failure by voltage sag alone, a criterion SOH takes no part in, and with-SOH transfer falls to 0.723 — a drop of about 0.17 (roughly 19%), with the with-SOH-over-without-SOH gap narrowing from +0.15 to +0.11. The residual signal under the voltage label (0.723 versus 0.611 without SOH) is consistent with genuine but weak health-state information: cells far below nominal capacity do fail sooner under almost any definition. Within a single dataset the distinction nearly vanishes — on NASA, under the SOH-defined label, the with-SOH model reaches pooled AUC 0.936, and even the sensors-only model tracks the label at 0.797 — because within one laboratory the sensor features already co-move with SOH. It is precisely that co-movement which dataset shift destroys, and with it the transfer illusion.

III-G. Same-Chemistry Cross-Laboratory Controls

If the collapse of Section III-C were a chemistry effect, transfer should survive when chemistry is held constant. Table 11 shows it does not. NASA→CALCE (both LCO, different laboratories and protocols) collapses without SOH just as LCO→LFP does, and Severson→Oxford (both LFP) reproduces the full pattern of the cross-chemistry shifts, including high with-SOH transfer and

Table 10: Failure-Definition Ablation at H=20 (XGBoost, ALL-LCO Source, Pooled AUC [CI]): Transfer With and Without SOH as a Feature, for Three Label Endpoints.

Label endpoint	Oxford		Severson	
	with SOH	no SOH	with SOH	no SOH
Combined (SOH or sag)	0.917 [0.878,0.965]	0.429 [0.279,0.597]	0.896 [0.812,0.995]	0.750 [0.692,0.835]
SOH only	0.995 [0.988,1.000]	0.464 [0.320,0.596]	0.985 [0.977,0.992]	0.759 [0.709,0.801]
Voltage sag only	—	—	0.723 [0.705,0.740]	0.611 [0.585,0.637]

Table 11: Same-Chemistry Cross-Laboratory Controls at H=20 (XGBoost, Pooled AUC [CI]). Full Feature Sets Exercise the Zero-Imputation Confound; Common Sets Restrict Both Datasets to the Shared Four-Feature Subset.

Pair	with SOH (full)	no SOH (full)	no SOH (common)
NASA→CALCE (LCO→LCO)	0.870 [0.799,0.966]	0.849 [0.753,0.985]	0.781 [0.708,0.964]
CALCE→NASA (LCO→LCO)	0.543 [0.356,0.677]	0.566 [0.427,0.678]	0.601 [0.496,0.703]
Severson→Oxford (LFP→LFP)	0.997 [0.991,1.000]	0.471 [0.466,0.477]	0.428 [0.422,0.435]
Oxford→Severson (LFP→LFP)	0.688 [0.605,0.788]	0.500 [0.500,0.500]	0.500 [0.500,0.500]

near-chance sensor-only transfer. The with-SOH numbers are again mirrored by the SOH distance rule. These controls tighten the paper’s central attribution: the failure to transfer is a property of *dataset shift* — different laboratories, protocols, and logging conventions — that chemistry change adds to but does not uniquely cause. Accordingly, we describe SOH as a *dataset- and chemistry-dependent* health-state signal, not a chemistry-specific one, and we do not claim chemistry is isolated as a causal variable.

III-H. Discrete-Time Hazard Model Versus Fixed-Horizon Classifiers

Table 12 compares the survival-product model of Section II-G against the fixed-horizon XGBoost classifier it is meant to complement. Within datasets the two formulations match to within a few AUC points at every horizon (Table 2); no formal equivalence test is run, so this is a point-estimate agreement, not a statistical indistinguishability claim. Under transfer the survival product inherits the same SOH dependence as everything else — the formulation of a model does not exempt its inputs from dataset shift — but it behaves differently in one structural respect: by construction $R(t, H)$ is monotone in H , whereas the four independent classifiers violate the corresponding requirement $p_{10} \leq p_{20} \leq p_{30} \leq p_{50}$ on 32–45% of within-dataset rows (raw scores; Table 13), rising to 95.8% (nasa-trained xgboost) on transferred Severson rows. For a hazard-flavored formulation the violations are

Table 12: Discrete-Time Hazard Model at H=20: Transfer Performance (Pooled AUC [CI]). The Fixed-Horizon XGBoost Comparator With SOH Is in Table 4.

Training	Model	Oxford	Severson
NASA	Hazard XGBoost	0.971 [0.953,0.991]	0.884 [0.786,0.998]
NASA	Hazard Logistic	0.500 [0.500,0.500]	0.607 [0.402,0.890]
CALCE	Hazard XGBoost	0.466 [0.301,0.578]	0.737 [0.711,0.774]
CALCE	Hazard Logistic	0.500 [0.500,0.500]	0.500 [0.500,0.500]
ALL LCO	Hazard XGBoost	0.314 [0.135,0.457]	0.866 [0.770,0.978]
ALL LCO	Hazard Logistic	0.500 [0.500,0.500]	0.916 [0.873,0.969]

not cosmetic: monotone cumulative risk is what makes “risk within 50 cycles” a coherent quantity to quote to an operator. The honest summary is that the discrete-time hazard model buys structural coherence and a single fit at no within-dataset accuracy cost, and pays the same SOH shortcut as every other model.

III-I. Failure of Calibration Transfer, and Its Partial Repair

A second, independent failure mode appears in the cross-dataset tables when calibrated outputs are required. Isotonic regression is fitted as a step function on the source distribution; when target scores shift, many target scores fall into one bin and receive identical calibrated probabilities, and pooled ranking degrades accordingly — in Table 4, isotonic calibration costs the tree ensembles up to 0.48 pooled AUC on Oxford and 0.31 on Severson even with SOH present. Platt’s smooth sigmoid degrades more gracefully, which is precisely the sense in which “Platt better preserves discrimination at comparable calibration error.” Fig. 2(b) shows the deeper problem: on the shifted target, *no* source-fitted calibrator lands near the diagonal, however it is fitted.

The recalibration arms of the companion study [14] ask whether a small labeled target sample repairs this. Arm A refits a calibrator on k labeled target cells while leaving the classifier’s scores untouched; Arm B continues training the classifier on them. On Severson without SOH, zero-shot ECE is 0.817; at $k=5$ labeled cells, isotonic and Platt reduce it to 0.044 and 0.045, while temperature scaling remains poorly calibrated (0.563). Calibration error is repaired, but discrimination is not: mean pooled AUC at $k=5$ stays near 0.61, because the calibrators create ties or reverse rankings on the

Table 13: Monotonicity Violations: Share of Scored Rows With $p_{10} \leq p_{20} \leq p_{30} \leq p_{50}$ Failing, on Raw Scores (Platt in Parentheses). The Survival Product Is Monotone by Construction (Rate 0).

Setting	Target	Model	Raw	Platt
within	calce	XGBoost	0.452	0.056
within	nasa	XGBoost	0.323	0.260
transfer (calce src)	oxford	LightGBM	0.641	0.004
transfer (calce src)	oxford	Random Forest	0.501	0.540
transfer (calce src)	oxford	XGBoost	0.565	0.003
transfer (nasa+calce src)	oxford	LightGBM	0.210	0.648
transfer (nasa+calce src)	oxford	Random Forest	0.335	0.305
transfer (nasa+calce src)	oxford	XGBoost	0.609	0.341
transfer (nasa src)	oxford	LightGBM	0.521	0.468
transfer (nasa src)	oxford	Random Forest	0.559	0.376
transfer (nasa src)	oxford	XGBoost	0.248	0.124
transfer (calce src)	severson	LightGBM	0.405	0.013
transfer (calce src)	severson	XGBoost	0.420	0.072
transfer (nasa src)	severson	LightGBM	0.582	0.537
transfer (nasa src)	severson	Random Forest	0.778	0.557
transfer (nasa src)	severson	XGBoost	0.958	0.923
—	—	Survival product	0.000	0.000

shifted target. Arm B is source-dependent: with NASA-trained XGBoost and $k=5$, Severson AUC rises from 0.620 to 0.854, overshooting the within-LCO reference ceiling (ratio 1.04, an artifact of the mismatched ceiling rather than genuine super-ceiling recovery), while CALCE-trained updates leave the source operating point essentially untouched (LCO-holdout retention 0.997 to 0.996 for CALCE-trained XGBoost) and ALL-LCO updates damage it substantially (see Table 15). Tables 14 and 15 summarize both arms.

III-J. Statistical Evidence for the SOH Ablation

Table 16 states the paper’s central comparison with the cell as the experimental unit: the paired cell-level bootstrap interval on $\Delta\text{AUC} = \text{AUC}(\text{with SOH}) - \text{AUC}(\text{without SOH})$, for the ALL-LCO source at every horizon, with the cycle-level DeLong p -value attached as secondary evidence. On Severson, ΔAUC reaches 0.15 [0.10, 0.23] (XGBoost, $H=20$) with the interval excluding

Table 14: Arm A Recalibration on Severson Without SOH ($H=20$): Values Average the Three LCO Sources and Available Tree Models; ECE Uses 10 Equal-Width Probability Bins. From the Released Recalibration Study Artifacts.

k	Zero ECE	Iso. ECE	Platt ECE	Temp. ECE	Iso. AUC	Platt AUC	Temp. AUC
5	0.817	0.044	0.045	0.563	0.616	0.599	0.669
10	0.817	0.010	0.010	0.561	0.634	0.646	0.651
20	0.817	0.013	0.013	0.560	0.634	0.655	0.648
40	0.817	0.015	0.015	0.561	0.637	0.657	0.653

Table 15: Arm B Recovery at $k = 5$ on Severson Without SOH, $H = 20$. Recovery Ratio Uses the Within-LCO Ceiling; Retention Is the LCO-Holdout AUC Before and After the Update. *Ratio above 1.0 is a ceiling-mismatch artifact, not super-ceiling recovery (see text).

Training	Model	Zero AUC	Arm B AUC	Recovery	DeLong p	LCO retention
NASA	XGBoost	0.620	0.854	+1.04*	$< 10^{-3}$	1.000 $\rightarrow$ 1.000
NASA	LightGBM	0.493	0.838	+0.88	$< 10^{-3}$	1.000 $\rightarrow$ 0.997
NASA	Random Forest	0.724	0.751	+0.19	$< 10^{-3}$	0.959 $\rightarrow$ 0.314
CALCE	XGBoost	0.686	0.796	+0.53	$< 10^{-3}$	0.997 $\rightarrow$ 0.996
ALL LCO	XGBoost	0.760	0.828	+0.47	$< 10^{-3}$	0.690 $\rightarrow$ 0.567

zero by a wide margin; on Oxford the point estimates are likewise positive for every model but the paired intervals are undefined with five cells, one contributing a single positive row, so Oxford contributes directional consistency only. The DeLong values, computed on pooled cycle-level predictions, underflow to zero throughout — and, as Section II-J explains, are inflated by within-cell correlation, which is exactly why they are demoted here to a consistency check and reported as $p < 10^{-16}$ (underflow) rather than interpreted. The scientific claim does not rest on tiny p -values; it rests on a large effect with a bootstrap interval that stays far from zero under cell-level resampling, on the 141-cell Severson target, at every horizon, across three model families, and — decisively — on the failure-definition and same-chemistry controls of Sections III-F and III-G.

III-K. Operational Cost Analysis

AUC treats all errors as equal; a BMS does not. The operational question is whether a decision threshold tuned on source data still protects on the target. We select the threshold on source out-of-fold scores with a 10% false-alarm budget — the operationally honest choice, since no target labels exist at deployment — and apply it unchanged to the target (Table 17).

Table 16: SOH Ablation Evidence (ALL-LCO Source): Pooled AUC With and Without SOH, Paired Cell-Level Bootstrap 95% CI on the Difference (Primary), and Cycle-Level DeLong p -Value (Secondary; Inflated by Within-Cell Correlation).

Target	Model	H	AUC w/ SOH	AUC w/o	Delta AUC [boot 95% CI]	DeLong p
Oxford	XGBoost	10	0.980	0.433	+0.547 (CI undefined: 5 cells)	$< 10^{-16}$ (underflow)
Oxford	XGBoost	20	0.917	0.429	+0.488 (CI undefined: 5 cells)	$< 10^{-16}$ (underflow)
Oxford	XGBoost	30	0.991	0.377	+0.614 (CI undefined: 5 cells)	$< 10^{-16}$ (underflow)
Oxford	XGBoost	50	0.932	0.580	+0.351 (CI undefined: 5 cells)	$< 10^{-16}$ (underflow)
Oxford	LightGBM	10	0.996	0.547	+0.449 (CI undefined: 5 cells)	$< 10^{-16}$ (underflow)
Oxford	LightGBM	20	0.971	0.332	+0.639 (CI undefined: 5 cells)	$< 10^{-16}$ (underflow)
Oxford	LightGBM	30	0.743	0.484	+0.259 (CI undefined: 5 cells)	$9.7e - 13$
Oxford	LightGBM	50	0.832	0.448	+0.384 (CI undefined: 5 cells)	$< 10^{-16}$ (underflow)
Oxford	Random Forest	10	0.978	0.626	+0.352 (CI undefined: 5 cells)	$< 10^{-16}$ (underflow)
Oxford	Random Forest	20	0.989	0.581	+0.409 (CI undefined: 5 cells)	$< 10^{-16}$ (underflow)
Oxford	Random Forest	30	0.971	0.462	+0.508 (CI undefined: 5 cells)	$< 10^{-16}$ (underflow)
Oxford	Random Forest	50	0.933	0.460	+0.473 (CI undefined: 5 cells)	$< 10^{-16}$ (underflow)
Severson	XGBoost	10	0.878	0.741	+0.137 [0.079, 0.249]	$< 10^{-16}$ (underflow)
Severson	XGBoost	20	0.896	0.750	+0.146 [0.097, 0.233]	$< 10^{-16}$ (underflow)
Severson	XGBoost	30	0.906	0.745	+0.162 [0.115, 0.237]	$< 10^{-16}$ (underflow)
Severson	XGBoost	50	0.917	0.736	+0.180 [0.134, 0.250]	$< 10^{-16}$ (underflow)
Severson	LightGBM	10	0.870	0.736	+0.134 [0.105, 0.200]	$< 10^{-16}$ (underflow)
Severson	LightGBM	20	0.882	0.718	+0.164 [0.127, 0.227]	$< 10^{-16}$ (underflow)
Severson	LightGBM	30	0.872	0.771	+0.102 [0.030, 0.230]	$< 10^{-16}$ (underflow)
Severson	LightGBM	50	0.899	0.744	+0.155 [0.112, 0.214]	$< 10^{-16}$ (underflow)
Severson	Random Forest	10	0.903	0.740	+0.163 [0.133, 0.225]	$< 10^{-16}$ (underflow)
Severson	Random Forest	20	0.889	0.746	+0.143 [0.106, 0.226]	$< 10^{-16}$ (underflow)
Severson	Random Forest	30	0.919	0.748	+0.171 [0.138, 0.232]	$< 10^{-16}$ (underflow)
Severson	Random Forest	50	0.921	0.746	+0.175 [0.114, 0.264]	$< 10^{-16}$ (underflow)

On Severson with SOH, the transferred XGBoost threshold misses 32% of target failures while holding false alarms to 4%. The same protocol fails for the other two ensembles on the same target with SOH: LightGBM alerts on 65% of negatives and Random Forest misses 100% of failures (raw scores) — threshold transfer fails for two of three ensembles even where pooled ranking transfers, so the XGBoost operating point must not be read as an ensemble-general result. Without SOH, the XGBoost miss rate more than doubles to 70%, and the false-alarm budget is now spent (10%) with nothing to show for it; the calibrated score types do not differ meaningfully, because under shift the ranking is the asset and the ranking is what collapsed. On Oxford, nearly every row is positive under the coarse pooled labels, so virtually any threshold alerts on almost every cycle — that target offers no room for a discriminating operating point at all (Section II-B). Fig. 5 shows the corresponding decision-curve view on Severson: with SOH the model buys net benefit across a wide threshold range; without SOH the curve hugs zero. Calibration, in other words, is a second-order concern in this regime: first the model must rank the target cells, which — per Sections III-C–III-G — only SOH lets it do.

Table 17: Operational Evaluation at $H=20$ (XGBoost, ALL-LCO Source): Decision Threshold Selected on Source Out-of-Fold Scores With a 10% False-Alarm Budget, Then Applied Unchanged to the Target. Reported Are the Target Miss Rate (FNR) and Target False-Alarm Rate (FPR).

Target	Features	raw		Platt		isotonic		temperature	
		FNR	FPR	FNR	FPR	FNR	FPR	FNR	FPR
Oxford	with SOH	0%	93%	0%	93%	0%	93%	0%	93%
Oxford	no SOH	0%	92%	0%	92%	0%	92%	0%	92%
Severson	with SOH	32%	4%	32%	4%	31%	6%	32%	4%
Severson	no SOH	70%	10%	70%	10%	67%	12%	70%	10%

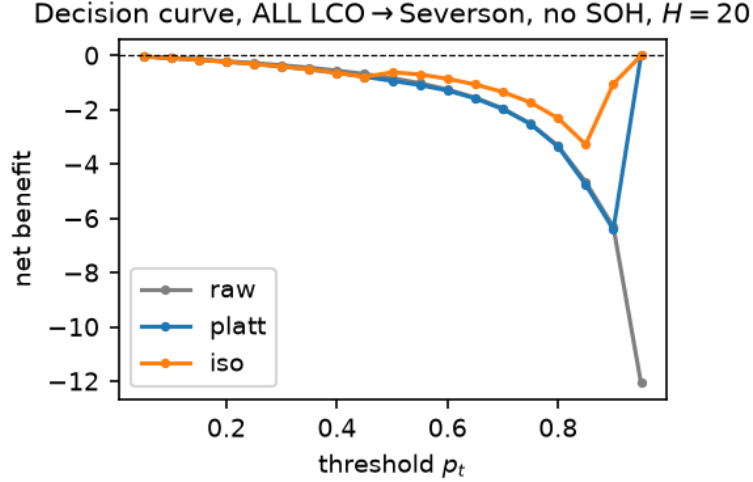


Figure 5: Decision-curve analysis (net benefit vs. threshold) for ALL-LCO $\rightarrow$ Severson without SOH at $H=20$: no score type produces a usable operating point once SOH is removed.

III-L. SHAP: The Shortcut Is Visible in the Model

TreeSHAP [15] gives a direct view of the mechanism. Figs. 6–8 show the attributions for the three tree models in NASA-to-Oxford transfer at $H=20$ with SOH kept: SOH dominates by a wide margin, cycle index a distant second, and the voltage, current, temperature, and duration features contribute little. Figs. 9–11 repeat the same comparisons with SOH removed, and every remaining feature collapses to near-zero SHAP spread — matching, in the model’s internal accounting, the quantitative collapse of Sections III-

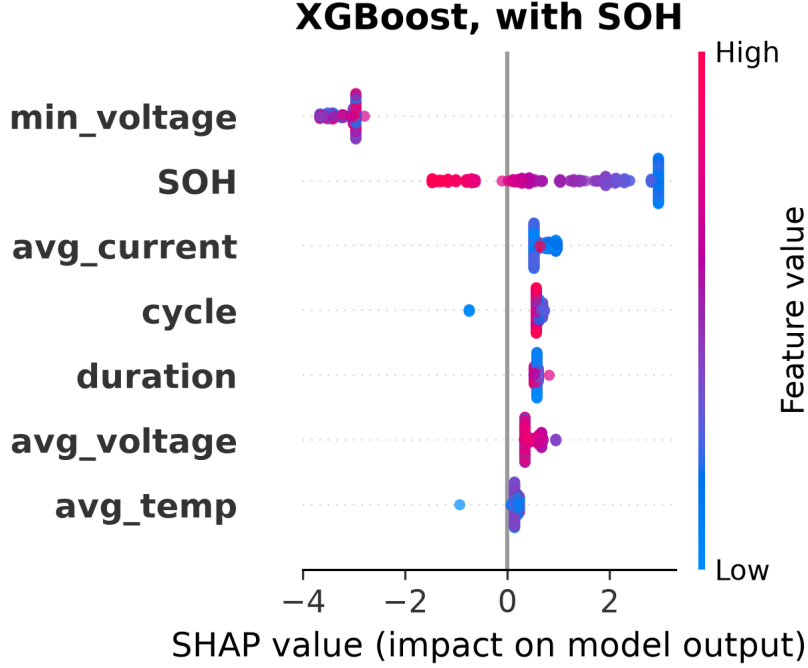


Figure 6: SHAP feature importance for XGBoost in NASA-to-Oxford transfer at H=20, with SOH. SOH dominates by a wide margin.

C-III-E and the sensor-inversion of Section III-D.

III-M. Embedded Deployment Validation

Three independent validation stages confirm that the C tree engine reproduces the Python training libraries through the output transforms of (9) and (10): a Python walker over the extracted trees, the same walker cross-checked on the full 1 028-row set, and the C engine run on-device on an ESP32-S3 against the library’s `predict_proba()`. Fig. 13 plots every one of the 1 028 on-device predictions against the Python values for all three models; the points fall on the identity line and the residual histograms are centered at zero. The measured on-device maximum absolute error is 2.4×10^{-7} for XGBoost, 1.1×10^{-6} for LightGBM, and 3.0×10^{-8} for Random Forest, all far inside the 10^{-5} tolerance.

Table 18 lists the models’ sizes, their NASA H=20 Platt AUC, the on-device validation error, and the measured inference time, and Fig. 12 plots the

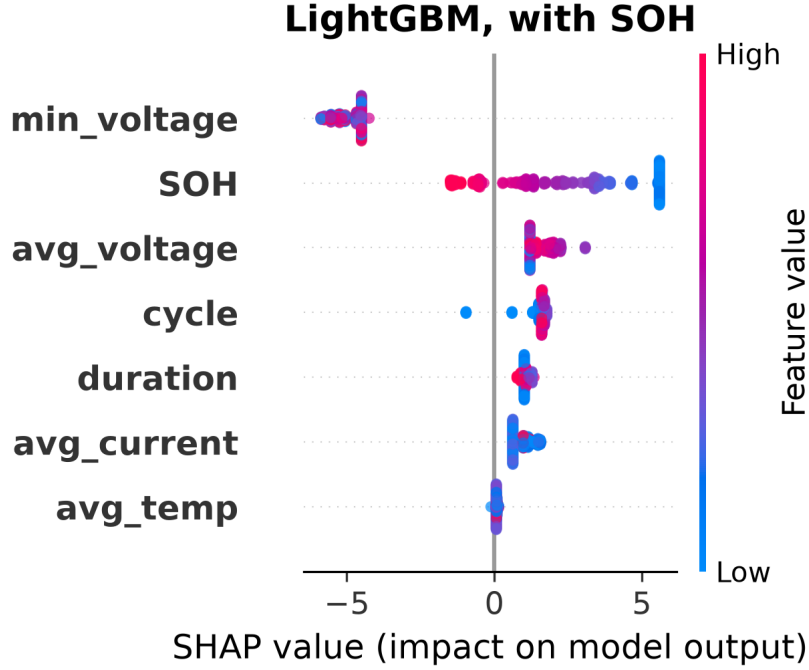


Figure 7: SHAP feature importance for LightGBM in NASA-to-Oxford transfer at $H=20$, with SOH. SOH dominates by a wide margin.

latency, the speed–accuracy trade-off, and the footprint. The three ensembles occupy about 372 kB in the flat binary. Measured single-row inference at 240 MHz depends on where the binary sits:

- from memory-mapped flash, 0.76 ms, 1.55 ms, and 2.36 ms for XGBoost, LightGBM, and Random Forest;
- from PSRAM, 0.50 ms, 0.93 ms, and 1.29 ms;
- with a single model held in internal SRAM, 0.32 ms, 0.43 ms, and 0.50 ms.

The full 372 kB binary does not fit internal SRAM, whose free heap is roughly 320 kB, but each individual model does (XGBoost is about 46 kB, LightGBM 98 kB, Random Forest 225 kB), and the firmware runs one model at a time, so the SRAM-resident numbers are the realistic per-inference cost. Both flash

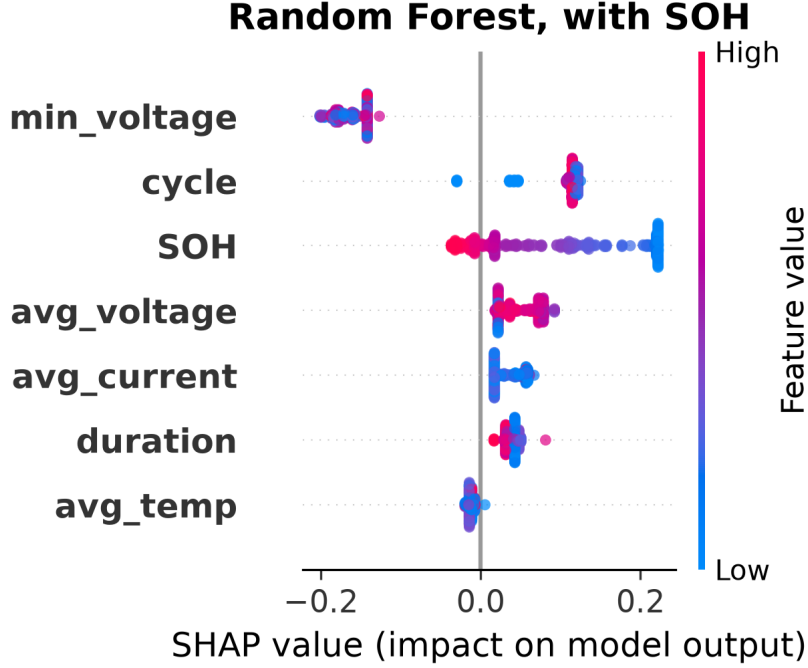


Figure 8: SHAP feature importance for Random Forest in NASA-to-Oxford transfer at H=20, with SOH. SOH dominates by a wide margin.

and PSRAM land well above the 150–250 μ s range projected at the outset, because the working set exceeds the ESP32-S3’s data cache and nearly every node read misses; PSRAM is about $1.5\text{--}1.9\times$ faster than flash for the same reason. The all-three PSRAM path still clears the 100 ms per-cycle budget by roughly $37\times$, and a single model from SRAM runs at roughly twice the originally projected 150–250 μ s. The deployment result is deliberately independent of the validity question: whatever the models are reading, on this dataset they can read it in under a millisecond on a \$12 microcontroller. The on-device outputs are raw ensemble probabilities through (9)–(10) with no calibration layer: the Platt and isotonic maps live off-device, and Section III-I shows source-fitted maps do not survive dataset shift, so a deployed system must revalidate or refit its calibrator on target-labeled cells rather than trust the laboratory calibration.

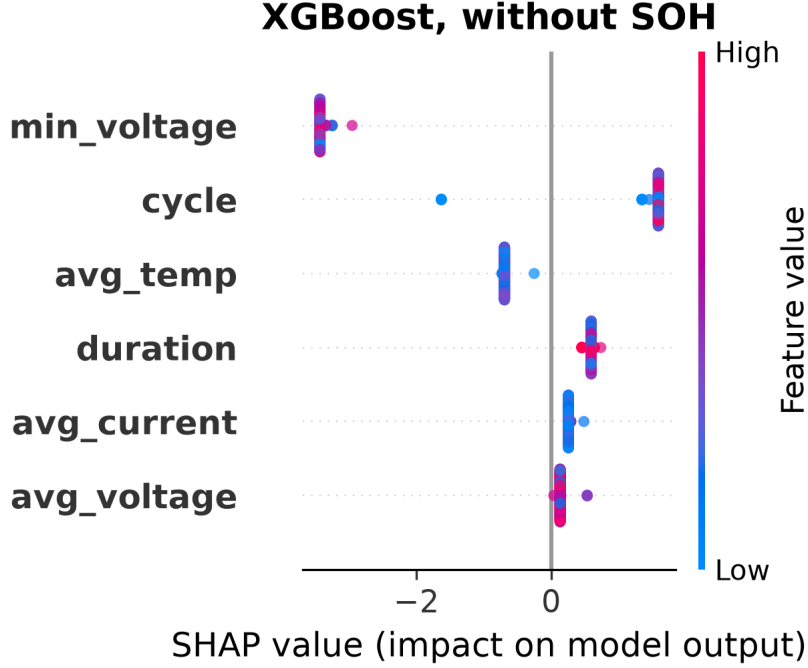


Figure 9: SHAP feature importance for XGBoost in NASA-to-Oxford transfer at H=20, without SOH. All features collapse to near-zero spread, so no transferable signal remains.

Table 18: Embedded Deployment: Model Size, Accuracy, On-Device Validation Error, and Measured Inference Time. Errors Are Maximum $|C \text{ engine} - \text{predict_proba()}|$ Over All 1 028 Rows. Timings Measured On-Device at 240 MHz With the Binary in Flash (PROGMEM), PSRAM, or a Single Model in Internal SRAM.

Model	Nodes	Max Err	Flash	PSRAM	SRAM (1 model)
XGBoost	3 272	2.4×10^{-7}	0.76 ms	0.50 ms	0.32 ms
LightGBM	7 000	1.1×10^{-6}	1.55 ms	0.93 ms	0.43 ms
Random Forest	16 064	3.0×10^{-8}	2.36 ms	1.29 ms	0.50 ms
All three	26 336	—	4.7 ms	2.7 ms	—

III-N. Replication on New Chemistries: NCA, NMC, and LFP Transfer Targets

The transfer claim is re-tested on 60 cells of NMC, NCA and LFP chemistry from two further laboratories (Battery Archive groups 5–8 of Section II-

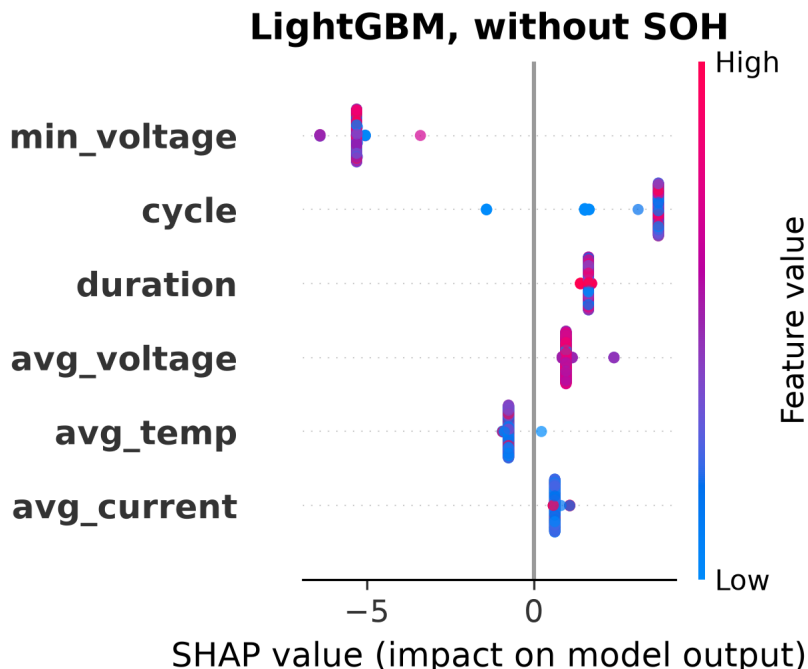


Figure 10: SHAP feature importance for LightGBM in NASA-to-Oxford transfer at $H=20$, without SOH. All features collapse to near-zero spread.

B), with the LCO training pool and protocol untouched — the new cells never enter model selection — and the experiment run on the same protocol version as every result above. Fig. 14 shows the post-cleaning capacity traces with the pre-registered audit decisions. Table 19 reports the NASA+CALCE $\rightarrow$ new-target transfer at $H=20$, with and without SOH, together with the paired cell-level bootstrap for the SOH ablation.

With SOH available, transfer from the LCO sources spans pooled AUC 0.77–1.00 across the new targets — the same upper-bound regime as Table 4. Removing SOH drops transfer to 0.36–1.00, and the paired per-cell bootstrap puts the with-minus-without difference at 0.00–0.64 AUC across the three ensembles — on the SNL groups the LCO-trained ensemble contributes nothing portable on NCA, NMC, or LFP chemistry from an unseen laboratory, exactly as it contributed nothing on the earlier LFP targets. The HNEI NMC group is the exception: full no-SOH transfer (cycle index included) stays

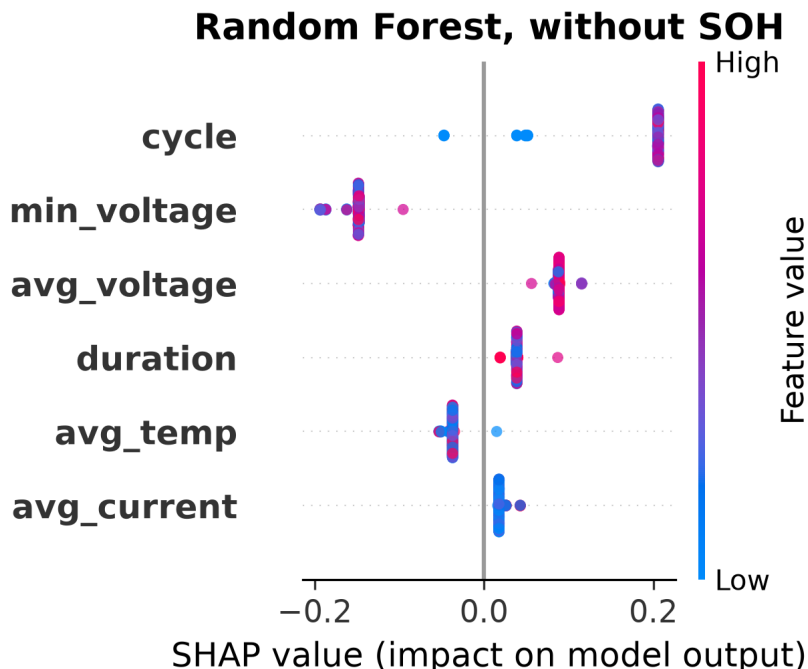


Figure 11: SHAP feature importance for Random Forest in NASA-to-Oxford transfer at H=20, without SOH. All features collapse to near-zero spread.

at 0.99 pooled there. All 15 HNEI cells cycle under one condition and one fixed charge protocol, so cycle-index and voltage cues survive the laboratory change; this is the dataset-shift mechanism of Section III-G operating in reverse — narrow, stereotyped protocols transfer, diverse ones do not — not chemistry-specific knowledge. The common-set column repeats the check on the unit-safe feature subset (Section II-C), controlling for the structurally missing current and temperature columns of the export: the SNL collapse is not an artifact of the zero-imputed columns — though only for the available telemetry, per the Limitations. On the new groups the voltage-sag endpoint never fires before the SOH endpoint (0 of 60 admitted cells), so the composite label is, effectively, the SOH criterion alone; the failure-definition ablation of Section III-F therefore cannot be re-run on these targets, and all with-SOH numbers here are to be read as label-proximity upper bounds, as defined in Section II-A.

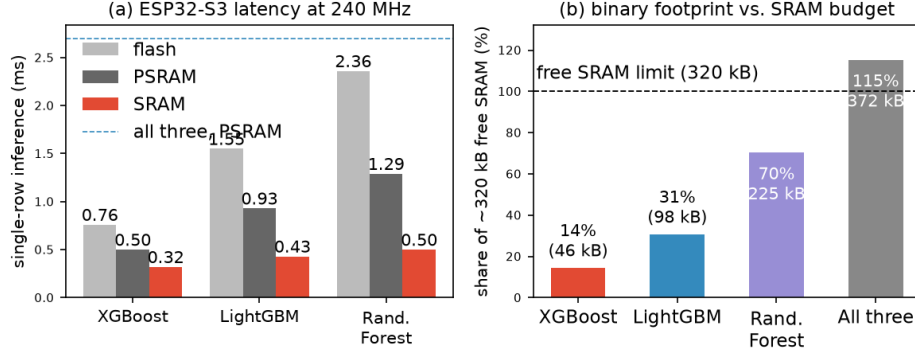


Figure 12: Deployment trade-offs: (a) measured single-row inference time per model at 240 MHz with the binary in memory-mapped flash, PSRAM, or a single model in internal SRAM (dashed line: all three models from PSRAM); (b) per-model and combined binary footprint as a share of the ESP32-S3’s roughly 320 kB of free internal SRAM — each model fits, the combined 372 kB binary does not.

Table 19: Transfer to New Chemistries at H=20 (NASA+CALCE Source). Pooled AUC With Cell-Level Bootstrap 95% CI; Δ Is the Paired Per-Cell Bootstrap Difference (With – Without SOH). New Groups Are Transfer Targets Only.

Target	Model	With SOH	Without SOH	Delta (paired)
HNEI NMC	XGBoost	0.993 [0.983,0.999]	0.989 [0.977,0.997]	+0.004
HNEI NMC	LightGBM	0.999 [0.998,0.999]	0.988 [0.974,0.996]	+0.011
HNEI NMC	Random Forest	0.999 [0.999,1.000]	0.996 [0.994,0.999]	+0.003
SNL NCA	XGBoost	0.972 [0.956,0.988]	0.878 [0.824,0.930]	+0.094
SNL NCA	LightGBM	0.985 [0.974,0.993]	0.850 [0.791,0.911]	+0.135
SNL NCA	Random Forest	0.983 [0.974,0.992]	0.911 [0.875,0.951]	+0.072
SNL NMC	XGBoost	0.772 [0.677,0.979]	0.709 [0.612,0.953]	+0.063
SNL NMC	LightGBM	0.847 [0.743,0.982]	0.693 [0.570,0.959]	+0.153
SNL NMC	Random Forest	0.858 [0.748,0.975]	0.742 [0.689,0.874]	+0.115
SNL LFP	XGBoost	0.998 [0.997,1.000]	0.496 [0.458,0.531]	+0.502
SNL LFP	LightGBM	0.999 [0.997,1.000]	0.359 [0.279,0.443]	+0.640
SNL LFP	Random Forest	0.999 [0.998,1.000]	0.666 [0.569,0.769]	+0.332

III-O. Operating-Condition Shift at Fixed Chemistry and Laboratory

The same-chemistry controls of Section III-G varied laboratory while holding chemistry fixed; here the complement is measured — hold chemistry *and* laboratory fixed and vary only the operating condition. For each SNL group (NCA, NMC, LFP), the test cell is chosen from an operating condition (temperature band, C-rate, or both) that contributes no training cell in

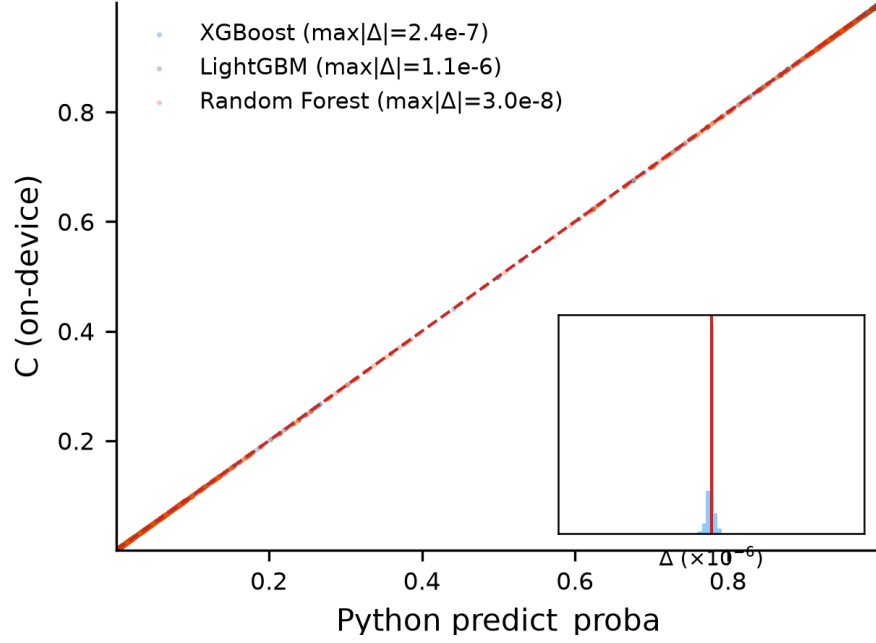


Figure 13: On-device C engine predictions versus Python `predict_proba()` over all 1 028 rows and three models (agreement scatter on the identity line, inset residual histogram of C minus Python in units of 10^{-6}). Maximum absolute errors are 2.4×10^{-7} (XGBoost), 1.1×10^{-6} (LightGBM), and 3.0×10^{-8} (Random Forest).

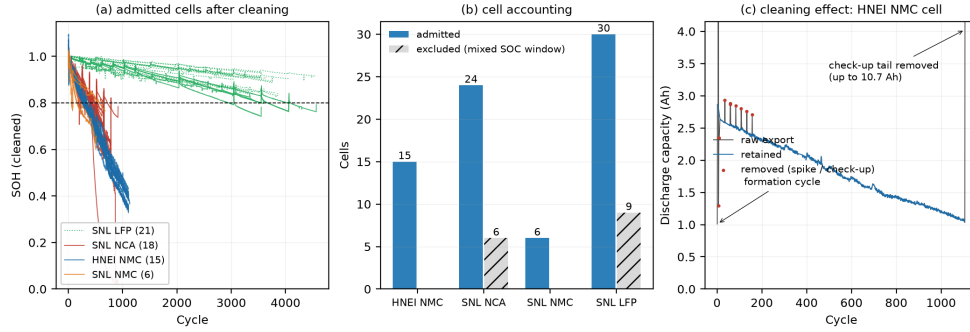


Figure 14: Data audit of the Battery Archive groups after the pre-registered cleaning rule (formation-cycle drop, Hampel despiking, level-shift truncation, voltage validity, minimum 20 usable cycles): (a) admitted-cell capacity traces with the 80% end-of-life criterion; (b) retained versus excluded cycles per group; (c) recorded exclusion reasons. Of 75 downloaded cells, 60 are admitted and 15 are excluded for exactly one recorded reason.

Table 20: Leave-Condition-Out Versus Mixed-Cell Folds at H=20 (XGBoost). Conditions Are Temperature Band, C-Rate, or Both; Metadata Never Enters the Features. Ref. Is the Ordinary Mixed-Cell AUC for the Same Feature Set (— Where Unstable or Not Run).

Group	Held-out axis	With SOH	Ref.	No SOH	Ref.
SNL NCA	temp $\times$ rate	0.989	0.985	0.894	0.928
SNL NCA	temperature	0.989	0.985	0.888	0.928
SNL NCA	C-rate	0.961	0.985	0.820	0.928
SNL NMC	temp $\times$ rate	0.847	—	0.666	—
SNL NMC	temperature	0.787	—	0.569	—
SNL NMC	C-rate	0.847	—	0.666	—
SNL LFP	temp $\times$ rate	0.991	0.994	—	—
SNL LFP	temperature	0.995	0.994	—	—
SNL LFP	C-rate	0.994	0.994	—	—

that fold; metadata parsed from cell identifiers provides the condition labels and is never used as a feature. Table 20 compares this leave-condition-out protocol with the ordinary mixed-cell folds on the same group.

Holding out an entire operating condition at fixed chemistry and laboratory costs little with SOH and more without it: on SNL-NCA the with-SOH pooled AUC moves only from 0.99 in ordinary mixed-cell folds to 0.99 under the joint holdout, and across all interpretable group-axis combinations the largest with-SOH drop is 0.02 AUC points — whereas the sensor-only model degrades before the with-SOH one does, falling to 0.89 on NCA (both axes) and 0.67 on NMC. Two caveats bound the claim’s strength. First, the mixed-fold raw reference on the 6-cell SNL-NMC group is itself unstable — with six cells the split composition dominates, and raw AUC there (0.65 with SOH) disagrees with the 0.87 that the same model reaches under the paper’s cell-disjoint outer folds — so its reference column is marked — and no drop is claimed for that group. Second, and most importantly, the distance-rule baseline is again undisturbed: its pooled AUC under condition holdout spans 0.886–0.999, essentially its mixed-fold value, so whatever the rule keys on is invariant to operating condition. Two conclusions follow. The residual within-laboratory signal that survives dataset shift is only mildly condition-dependent — chemistry and laboratory shift, not protocol condition, carry the larger penalty — and no competing shortcut of comparable size hides in temperature or C-rate: with-SOH scores remain well above their

no-SOH counterparts under condition holdout, so the SOH coordinate, not an operating-condition coordinate, is the portable quantity — portable for the definitional reason quantified in Section III-F. The HNEI group is omitted because its cells cycle under a single condition (25 °C, mixed fixed discharge C-rates under one charge protocol), leaving no condition axis to hold out. SNL LFP was run with the with-SOH and common no-SOH feature sets only, so its full-feature no-SOH column reads — by design, not by omission.

III-P. Key Findings

- Within-distribution failure-risk models are reliable (fold-mean AUC 0.90–0.96 across tree ensembles, Platt-calibrated) and run on-device in under a millisecond per single SRAM-resident model.
- Cross-dataset transfer collapses without SOH: on Severson from up to 0.90 with SOH to 0.53–0.75 without, and an unfitted SOH distance rule matches the tuned ensembles wherever they transfer.
- With-SOH numbers are diagnostic-proximity upper bounds, not pure prognosis: the label window includes the current cycle, and prognosis-only evaluation on rows scored while SOH is still above 0.80 gives 0.838 [0.760, 0.991] on Severson and 0.73–0.99 on the new targets (end of Section III-C).
- Dataset and laboratory shift govern the collapse at least as much as chemistry; the single-condition HNEI exception and the leave-condition-out controls show narrow protocols transfer while diverse ones do not, and no competing temperature or C-rate shortcut exists in the available, partially imputed telemetry (telemetry caveat, per the Limitations).
- Source-fitted calibration does not transfer: target recalibration repairs ECE but not ranking, and transferred decision thresholds fail for two of three ensembles even with SOH.

IV. Discussion, Analysis, and Novelty

IV-A. The SOH Shortcut, Precisely Stated

The evidence assembled above supports a precise statement — more precise, and more useful, than the earlier version of this study offered. When a

failure-risk model trained on one battery dataset is evaluated on another, its apparent transfer performance is dominated by the SOH coordinate; much of that dominance is definitional rather than learned, because SOH enters the failure label itself, and the failure-definition ablation of Section III-F measures how much: relabeling failure by voltage sag alone removes most of the with-SOH transfer advantage on Severson. What remains is a weak, real health-state signal — a cell far below nominal capacity is genuinely closer to failure under any reasonable definition. Everything else the sensors measure is dataset-specific: the same-chemistry controls show the collapse is caused by dataset shift at least as much as by chemistry, and the factorial ablation shows no non-SOH feature group carries usable cross-dataset signal. The replication of Section III-N extends each of these statements to two chemistries and two laboratories that played no part in the protocol’s development: the with-SOH transfer range 0.77–1.00 and its drop to 0.36–0.91 without SOH outside the single-condition HNEI group (paired deltas 0.06–0.64), and the absence of a competing condition shortcut (Section III-O), are confirmations measured on data the pipeline never had access to when it was designed. “SOH is a chemistry-specific lookup key,” the earlier framing, claimed more than the design could identify; the defensible claim is that apparent cross-dataset, cross-chemistry transferability is largely a health-state shortcut whose size this paper measures with label-, feature-, and dataset-level controls.

IV-B. Implications for Battery Management Practitioners

Four practical points follow:

- **1)** Within-chemistry, within-laboratory failure-risk models built from standard charge–discharge features are reliable — fold-mean AUC of 0.89 and above with Platt calibration — and run on a \$12 microcontroller in under a millisecond per single SRAM-resident model (slower from flash or PSRAM; Section III-M). Deploy them where the training distribution holds; the reliability extends to NMC, NCA and LFP chemistry from further laboratories (Section III-N), but always under cell-disjoint evaluation with conditions the model has seen.
- **2)** Do not transfer models across datasets — or across chemistries, or laboratories — on the strength of an AUC table; the collapse measured on the new-chemistry targets (Section III-N) is the same one measured within LFP, now confirmed outside the datasets the protocol was built

on, with the narrow-protocol HNEI exception noted there. If SOH is available at the target, an unfitted threshold rule is the correct first baseline, and any ensemble that cannot beat it has learned nothing portable; if SOH is unavailable, expect chance-level or inverted performance unless the target protocol is as narrow and stereotyped as the source.

- **3)** Quote risks from a formulation whose horizons are coherent. Four independent classifiers can and do assign “risk within 10 cycles” above “risk within 50 cycles”; the survival-product formulation of Section II-G removes that incoherence at no measured accuracy cost.
- **4)** Under shift, prefer raw or Platt-scaled scores over isotonic, expect calibration to require labeled target data (5–10 cells suffice for ECE, but they do not restore discrimination), and choose decision thresholds on source data with an explicit cost model — then verify, because Section III-K shows the operating point itself does not survive the shift without SOH.

IV-C. Limitations

The honesty of this paper has limits of its own. Oxford contributes only five cells — one with a single positive row, so per-cell means are effectively four-cell statistics — at a 100-cycle logging resolution that makes its labels horizon-invariant, so it functions as a pooled-detection target; every Oxford claim is re-verified on the 141-cell Severson set. CALCE contributes seven cells, so its leave-one-cell-out folds are few and its uncertainty is correspondingly wide. The new Battery Archive groups sharpen the chemistry claim but not the condition claim: the SNL NCA and NMC groups span a handful of temperature–rate conditions with as few as three cells per condition, so the leave-condition-out intervals of Section III-O are correspondingly wide, and the SNL LFP group, whose 21 cells are mostly censored within the recorded window and carry 0.031 positive-label prevalence at $H=20$, contributes supporting rather than headline evidence. The Battery Archive export provides capacity and voltage summaries but no per-cycle current or temperature columns; for those groups the affected features are zero-imputed like CALCE’s, and conclusions rest on the common unit-safe feature set rather than on the columns the export cannot supply. The sensor-feature collapse on the new targets is therefore a statement about the available, partially

imputed telemetry: it cannot rule out that richer per-cycle measurements might carry portable degradation information. The GRU is small and evaluated at three seeds, not thirty; its instability under shift is itself a finding, but wider sequence models trained on far more cells — let alone pretrained representations, transformers, or state-space models — might behave differently — nothing in this paper convicts the architecture, only this instance of it. The GRU was not re-evaluated on the Battery Archive targets, so the sequence-model results of Sections III-C and III-J rest on the LFP transfer experiments alone while Sections III-N and III-O cover the tree ensembles. The hazard model uses measured features at scored cycles; a strictly causal deployment reading must remember that post-failure rows are trivially positive for all formulations. The cost analysis fixes a 20:1 ratio and selects thresholds on source data; other ratios shift the operating point but not the conclusion that no usable operating point exists without SOH. Finally, the study’s transfer experiments remain one-directional (LCO-trained sources); the new groups are used as targets and controls only, and a controlled same-laboratory cross-chemistry experiment — which the Battery Archive groups only approximate, since chemistry there co-varies with protocol — would still be needed to isolate chemistry as a causal variable.

V. Conclusion

This study set out to widen a multi-horizon failure-risk classification framework and found that the widening itself demanded a validity analysis. Within dataset, the framework is sound: fold-mean AUC of 0.90–0.96 across three tree ensembles under cell-disjoint evaluation with leakage-free calibration, a discrete-time hazard formulation that matches the fixed-horizon classifiers while keeping cumulative risks monotone in the horizon, and a 372 kB embedded deployment that reproduces the training libraries to within 1.1×10^{-6} on a \$12 ESP32-S3.

Across datasets, the apparent transferability is mostly a State-of-Health shortcut. A distance-to-threshold rule on SOH alone reproduces the tuned ensembles’ transfer performance to within the cell-level bootstrap intervals; removing SOH collapses every model class toward or below chance, with sensor features sometimes inverting; relabeling failure by voltage sag alone — a criterion SOH does not define — removes most of the with-SOH advantage; and same-chemistry cross-laboratory controls show the collapse is caused by dataset shift at least as much as by chemistry. The supported conclusion

is deliberately narrower than “SOH is chemistry-specific”: much of the observed cross-dataset discrimination is attributable to SOH, partly because SOH defines the failure criterion itself, rather than to degradation information in the remaining measured features. Crucially, these statements are no longer confined to the datasets they were derived on: on 60 NMC, NCA and LFP cells from two further laboratories, admitted under a cleaning rule fixed before any model ran, with-SOH transfer spans 0.77–1.00 and drops to 0.36–0.91 without SOH outside the single-condition HNEI group (paired deltas 0.06–0.64), and holding out entire operating conditions while chemistry and laboratory stay fixed costs the sensor-only model up to 0.11 AUC points while the distance rule stays above 0.886, so the shortcut reproduces under laboratory and chemistry change, and no comparable shortcut hides in temperature or C-rate — a pre-registered replication, not a post-hoc fit.

Calibration behaves as the validity story predicts: fitted on leakage-free out-of-fold scores, Platt preserves discrimination best at comparable calibration error, isotonic’s tied step function destroys precision–recall and fails outright under shift, labeled-target recalibration repairs calibration error but not discrimination, and no score type yields a usable cost-based operating point on the shifted target without SOH. Statistical evidence is carried by cell-level bootstrap intervals, with cycle-level significance tests demoted to a consistency check under an explicit pseudoreplication caveat.

Future work, in order of directness: an SOH-free failure definition (voltage- or impedance-based endpoints) as the primary target for cross-dataset models, re-tested on the Battery Archive groups once per-cycle voltage exports support the sag criterion there; same-laboratory, cross-chemistry aging campaigns that isolate chemistry; learned representations trained for dataset invariance rather than chemistry invariance; cell-level hierarchical calibration that propagates uncertainty from cells to risk statements; and multi-seed, multi-architecture sequence-model studies on the scale where sequence models can plausibly generalize.

Acknowledgment

This work received no external funding. The authors declare no conflicts of interest. The source code, data, trained models, firmware, and validation pipeline are publicly available at <https://github.com/touhidsiddiqueeraj-bit/Multi-Horizon-Hazard-Models-for-Battery-Failure-Prediction>.

References

- [1] H. Meng and Y. Li, “A review on prognostics and health management (PHM) methods of lithium-ion batteries,” *Renew. Sustain. Energy Rev.*, vol. 116, Art. no. 109405, 2019.
- [2] D. Roman, S. Saxena, V. Robu, et al., “Machine learning pipeline for battery state-of-health estimation,” *Nat. Mach. Intell.*, vol. 3, no. 4, pp. 447–456, 2021.
- [3] V. Sulzer, P. Mohtat, A. Aitio, et al., “The challenge and opportunity of battery lifetime prediction from field data,” *Joule*, vol. 5, no. 8, pp. 1934–1955, 2021.
- [4] L. Breiman, “Random forests,” *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, 2001.
- [5] T. Chen and C. Guestrin, “XGBoost: A scalable tree boosting system,” in *Proc. ACM SIGKDD*, 2016, pp. 785–794.
- [6] G. Ke, Q. Meng, T. Finley, et al., “LightGBM: A highly efficient gradient boosting decision tree,” in *Proc. NeurIPS*, vol. 30, 2017, pp. 3146–3154.
- [7] T. A. Shikdar and H. Laaksonen, “Learning when not to use a battery: Multihorizon failure intelligence,” *Int. Trans. Electr. Energy Syst.*, vol. 36, Art. no. 6000810, 2026.
- [8] S. Sahoo, K. S. Hariharan, S. Agarwal, et al., “Transfer learning based generalized framework for state of health estimation of Li-ion cells,” *Sci. Rep.*, vol. 12, Art. no. 13173, 2022.
- [9] J. Lu, R. Xiong, J. Tian, C. Wang, and F. Sun, “Deep learning to estimate lithium-ion battery state of health without additional degradation experiments,” *Nat. Commun.*, vol. 14, Art. no. 2760, 2023.

- [10] J. C. Platt, “Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods,” in *Adv. Large Margin Classifiers*, MIT Press, 1999, pp. 61–74.
- [11] B. Zadrozny and C. Elkan, “Transforming classifier scores into accurate multiclass probability estimates,” in *Proc. ACM SIGKDD*, 2002, pp. 694–699.
- [12] A. Niculescu-Mizil and R. Caruana, “Predicting good probabilities with supervised learning,” in *Proc. ICML*, 2005, pp. 625–632.
- [13] L. Huang, J. Zhao, B. Zhu, H. Chen, and S. K. L. M. v. Broucke, “An experimental investigation of calibration techniques for imbalanced data,” *IEEE Access*, vol. 8, pp. 127343–127352, 2020.
- [14] T. A. Shikdar and H. Laaksonen, “How far does the model hold? Robustness of multihorizon battery failure intelligence under realistic operating conditions,” submitted for publication, 2026.
- [15] S. M. Lundberg, G. G. Erion, H. Chen, et al., “From local explanations to global understanding with explainable AI for trees,” *Nat. Mach. Intell.*, vol. 2, no. 1, pp. 56–67, 2020.
- [16] P. Grzesik and D. Mrozek, “Combining machine learning and edge computing: Opportunities, challenges, platforms, frameworks, and use cases,” *Electronics*, vol. 13, no. 3, Art. no. 640, 2024.
- [17] C. Gupta and A. Ramdas, “Online Platt scaling with calibrating,” in *Proc. ICML*, PMLR 202, 2023, pp. 12182–12204.
- [18] B. Saha and K. Goebel, “Battery data set,” NASA Ames Prognostics Data Repository, 2007.
- [19] CALCE Battery Research Group, “Battery aging datasets,” Univ. Maryland, 2023.
- [20] C. R. Birkel and D. A. Howey, “Oxford battery degradation dataset 1,” Univ. Oxford, 2017.
- [21] K. A. Severson, P. M. Attia, N. Jin, et al., “Data-driven prediction of battery cycle life before capacity degradation,” *Nature Energy*, vol. 4, pp. 383–391, 2019.

- [22] P. M. Attia, A. Grover, N. Jin, et al., “Closed-loop optimization of fast-charging protocols for batteries with machine learning,” *Nature*, vol. 578, no. 7795, pp. 397–402, 2020.
- [23] Battery Archive, “Open battery degradation testing data,” 2023. [Online]. Available: <https://batteryarchive.org>. Consolidated cell cycling records from the Hawaii Natural Energy Institute and Sandia National Laboratories.
- [24] Y. Preger, H. C. Barkholtz, A. Fresquez, et al., “Degradation of commercial lithium-ion cells as a function of chemistry and cycling conditions,” *J. Electrochem. Soc.*, vol. 167, no. 12, Art. no. 120532, 2020.
- [25] E. R. DeLong, D. M. DeLong, and D. L. Clarke-Pearson, “Comparing the areas under two or more correlated receiver operating characteristic curves: A nonparametric approach,” *Biometrics*, vol. 44, no. 3, pp. 837–845, 1988.